



# Surgical robotics today

Who are all these companies &  
How did they get here?



Dwight Meglan

July 2026 Update · The commercial surgical robotics landscape  
**Companion to the Surgical AI Overview (July 2026)**

*All information discussed is derived from publicly available sources*

# Why me for this subject?

30+ years building surgical robots and simulators, and tracking everyone else's

- **Left Mayo tenure track 30+ years ago to stay an engineer**

Building surgical simulators since 1994; surgical robots since 2000

- **Part of many investment & acquisition diligence efforts**

Surgical robotics and adjacent: robotic systems, components and enablers; advanced visualization; machine learning

- **20+ years tracking commercial robots & simulators**

I work at the system level, from idea to prototype

## 15 COMMERCIAL SURGICAL ROBOTS

Involvement across specialties:

- 4 Abdominal / thoracic
- 4 Endovascular
- 3 Endoluminal
- 2 Cardiac
- 1 Ophthalmic
- 1 Vascular access

# Why am I doing this?

A hobby that turned out to be useful, everything is a remix

- **I started collecting this information as a hobby**

Knowing how similarly focused robots could have diverse solutions was useful for my design work (everything is a remix)

- **As the collection grew, others might find it useful**

This deck is an overview of the field touching on the diversity of solutions for similar needs, e.g. surgeon consoles

- **I do deep dives for my own startup work and consulting**

## NOTE

This talk is largely mechanics focused, but I typically work at the system level

- I work at the idea-to-prototype phase, i.e. typically much earlier than Steve Bell :-)

# This talk...

A scope contract: breadth over depth, commercial over academic

## IS

- A quick run through the breadth of commercial efforts
- An update on what has changed in the past ~6 months
- Examples of how this information provides insight
- Robots that act upon the body in some manner
- A broad view of 'commercial', even patents alone

## IS NOT

- By any means exhaustive
- Delving into the myriad academic efforts
- Covering rehabilitation or assistive robots

# Surgical robotics today: update notes, July 2026

What changed since the January 2026 edition

- **45 additions since January**

Roster now 760 robots / 663 companies: 7 orthopedic, 5 abdominal/thoracic, 5 percutaneous, 4 endovascular, 3 dental... 30 of the 45 are Chinese, mostly EPO patent-scan finds, not press

- **Full roster re-verification, 1-2 July**

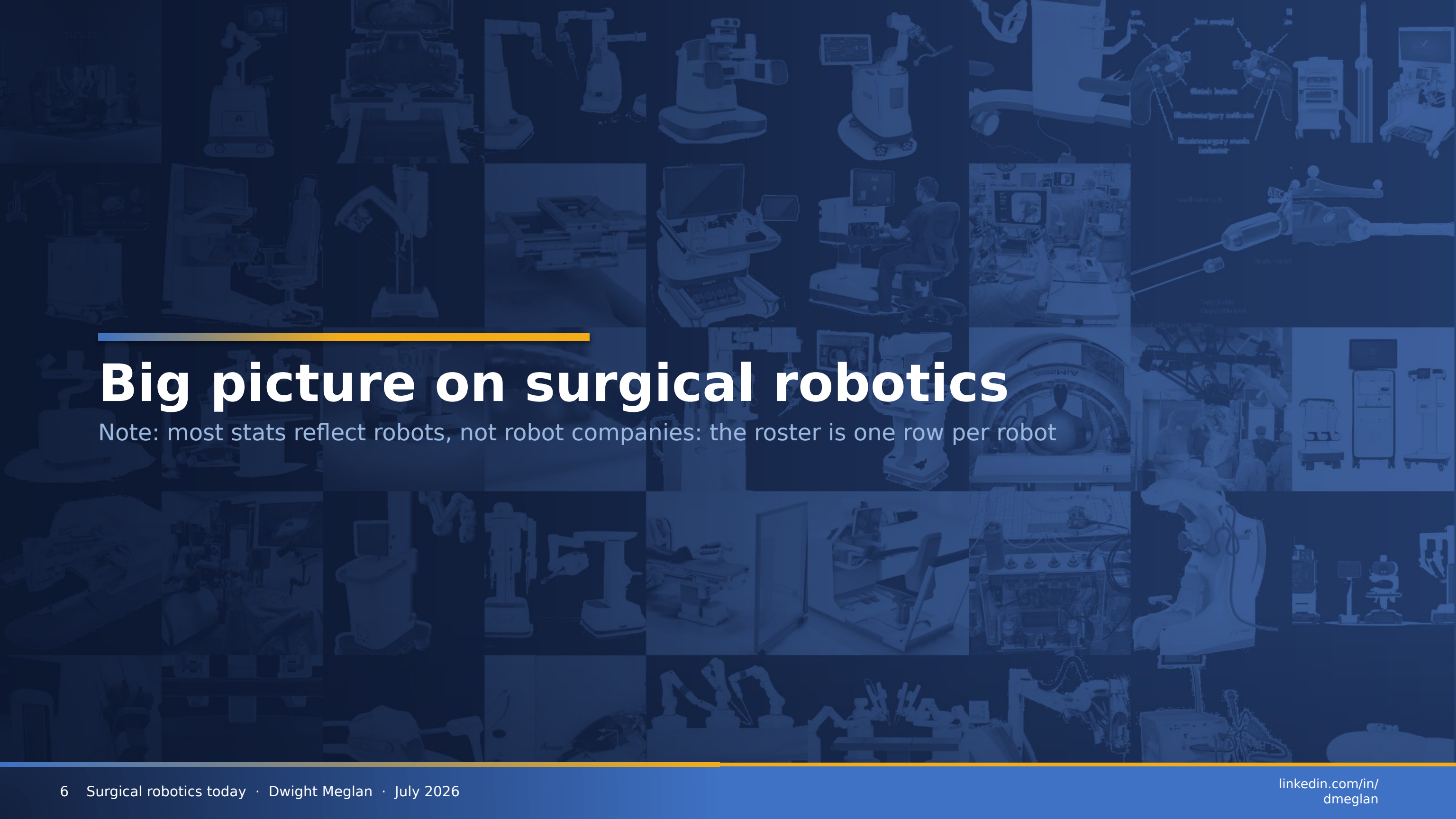
532 field corrections, 121 new milestones, duplicates merged; quietly-approved systems re-checked, On Market count: 86 in January → 239 now

- **Funding data added for every robot**

Stage, total raised, lead investors, valuation and most-recent-milestone columns now populated throughout

## ALSO NEW IN THIS EDITION

- Funding & investor data throughout
- 2026 flameouts & exits section
- Founding-year (vs first-evidence) view
- All charts regenerated from the master roster
- Live companion dashboard, see the end of the deck



# Big picture on surgical robotics

Note: most stats reflect robots, not robot companies: the roster is one row per robot

# Surgical robotics is a busy place

316 review articles in 2020–2022 per the refreshed counts; a sampling of 2022 titles below

## HISTORY & OVERVIEWS

A short history of robotic surgery · Past, present and future of surgical robotics · An introductory review of robotically assisted surgical systems

## IMAGING & GUIDANCE

Image-Guided Interventional Robotics: Lost in Translation? · Robot-Assisted Medical Imaging: A Review · MRI-Guided Robot-Assisted Surgery and Interventions

## SPECIALTY-SPECIFIC

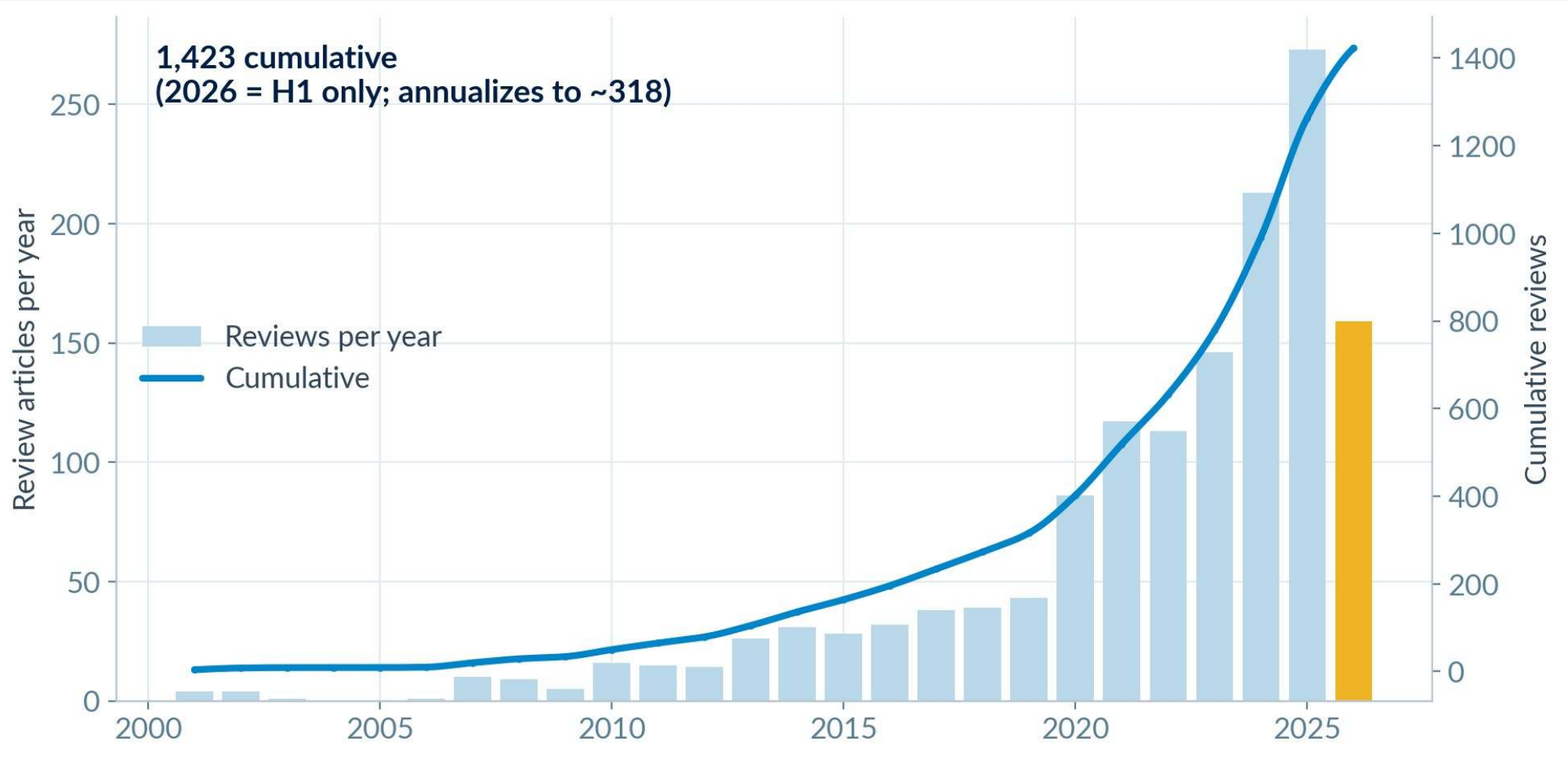
Robotics in neurosurgery · Intraocular Robotic Surgical Systems · Robot-assisted techniques in vascular and endovascular surgery · Robotic Systems in Radiotherapy

## EMERGING TECH

Magnetically Actuated Medical Robots · Surgical Robotics in the Data Age · among others published in 2022

# Surgical robotics is a busy place

Review articles per year, Google Scholar, refreshed 7 Jul 2026; 2026 = first half, actual



**1,423 REVIEWS**

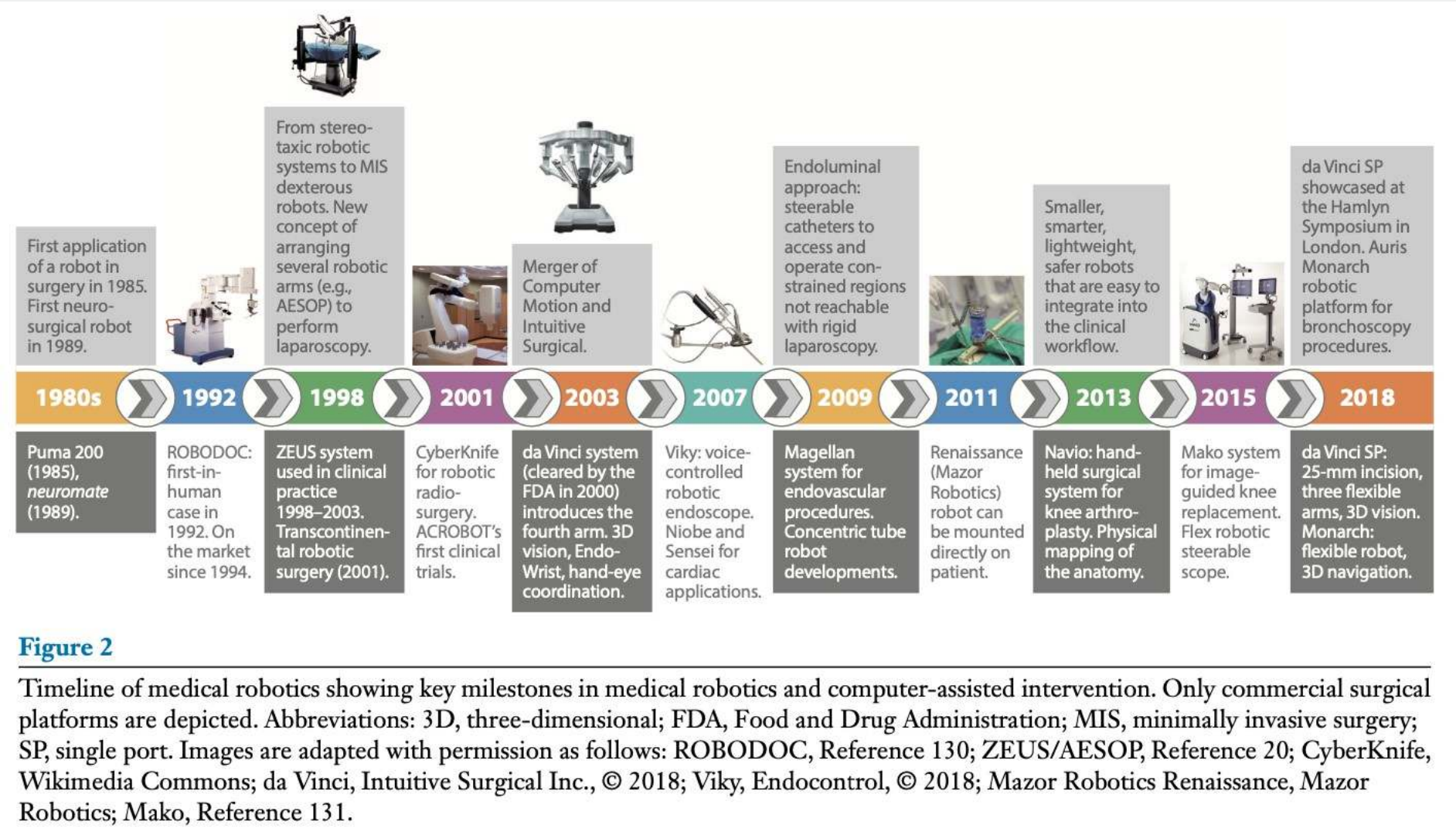
**cumulative since 2001**

273 in 2025; 159 in H1 2026, annualizing to roughly 318

**Same Google Scholar query as prior editions; counts refreshed 7 Jul 2026**

# Where it came from

J Troccaz et al., Frontiers of Medical Robotics, Annu. Rev. Biomed. Eng. 21:193-218, 2019



J Troccaz et al., Frontiers of Medical Robotics: From Concept to Systems to Clinical Translation

# Where it came from

knowablemagazine.org, Handing the surgeon's scalpel to a robot (2022)

## Timeline of progress in surgical robots

- 

**1970s** NASA explores telesurgery.
- 

**1983** First robot in the surgical suite, used to move patient's limbs into position during orthopedic surgery.
- 

**1985** First instance of a surgical robot, used during brain biopsy to avoid errors from hand tremors.
- 

**1985** First laparoscopic procedure.
- 

**1990s** Laparoscopic surgery becomes more common, many novel procedures carried out and many new systems invented.
- 

**1991** First telepresence system, means surgeons don't need to be directly next to patient.
- 

**1992** First robot with task autonomy gets FDA approval. Used in hip replacement, ROBODOC can carry out certain pre-programmed tasks autonomously once initiated by a human.
- 

**2000s** Surgical robots for general use become more common.
- 

**2000** Introduction of da Vinci, the first robotic system for general and laparoscopic surgery in humans.
- 

**2001** Doctors in New York City use telesurgery to operate on a patient in France.
- 

**2001** First robot with conditional autonomy gets FDA approval. Used in radiation therapy, CyberKnife can perform procedure autonomously.
- 

**2001 – present** Refinement of existing surgical robots continues to expand, robots become smaller, lighter, easier to use, and offer more assistance features for surgeons.
- 

**2009** Steerable catheters to access and operate constrained regions not reachable with rigid laparoscopy.
- 

**2022** STAR robot successfully sutures severed pig intestine.
- 

**The future** Surgical robots gradually acquire increasing autonomy.

SOURCE: REPORTING BY J. GAINES

KNOWABLE MAGAZINE

<https://knowablemagazine.org/article/technology/2022/handing-surgeon-scalpel-to-robot>

# Commercial milestones

T Ginoya et al., A Historical Review of Medical Robotic Platforms, J. Robotics 2021

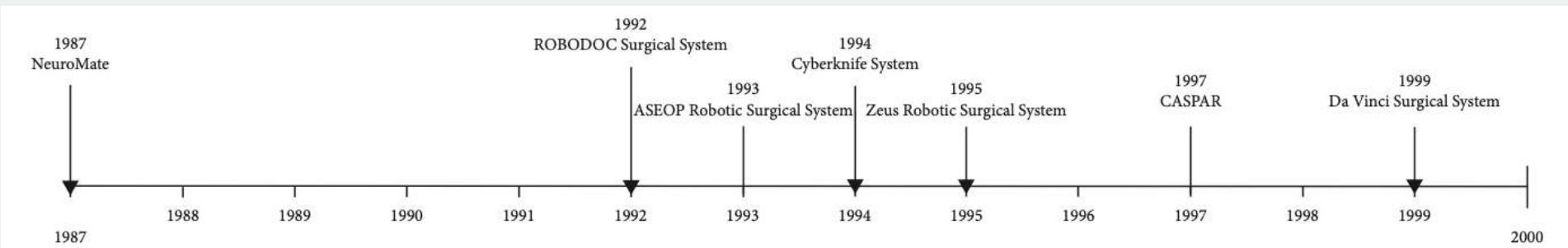


FIGURE 1: Decade I: timeline of robotic systems that founded and created the base for usage of robots in the medical field.

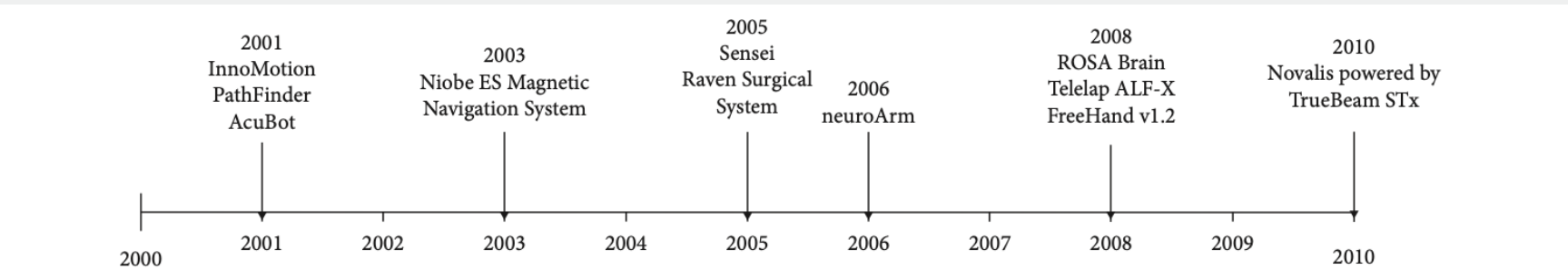


FIGURE 2: Decade II–timeline of robotic systems that signify the progress of the development curve of Robots in medical application.

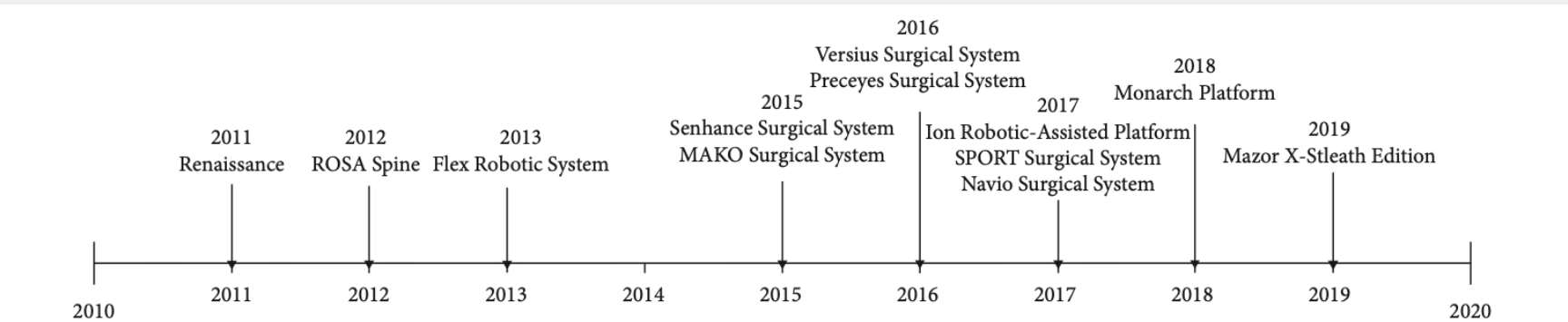


FIGURE 3: Decade III–timeline of robotic systems that show the recent advancement in the field of medical robots based on the advancement of technology and effort of three decades of research in this field.

T Ginoya et al., A Historical Review of Medical Robotic Platforms  
Journal of Robotics Volume 2021, Article ID 6640031,  
13 pages <https://doi.org/10.1155/2021/6640031>



# Commercial surgical robotics trends

Five parts: the roster today · how fast it grew · where · specialty waves · the endovascular case.  
Stats reflect robots, not companies.

# Lots of bots, mostly on the way

Roster · the field at a glance; there are more: stealth and non-US companies evade detection

760

robots tracked

663

companies

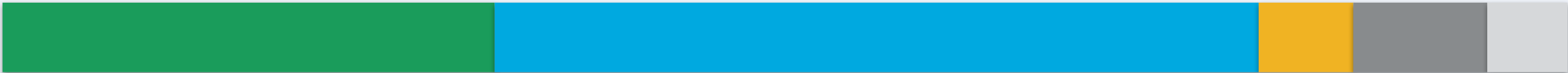
239

on market

67

multi-robot companies

## STATUS OF THE 760 ROBOTS

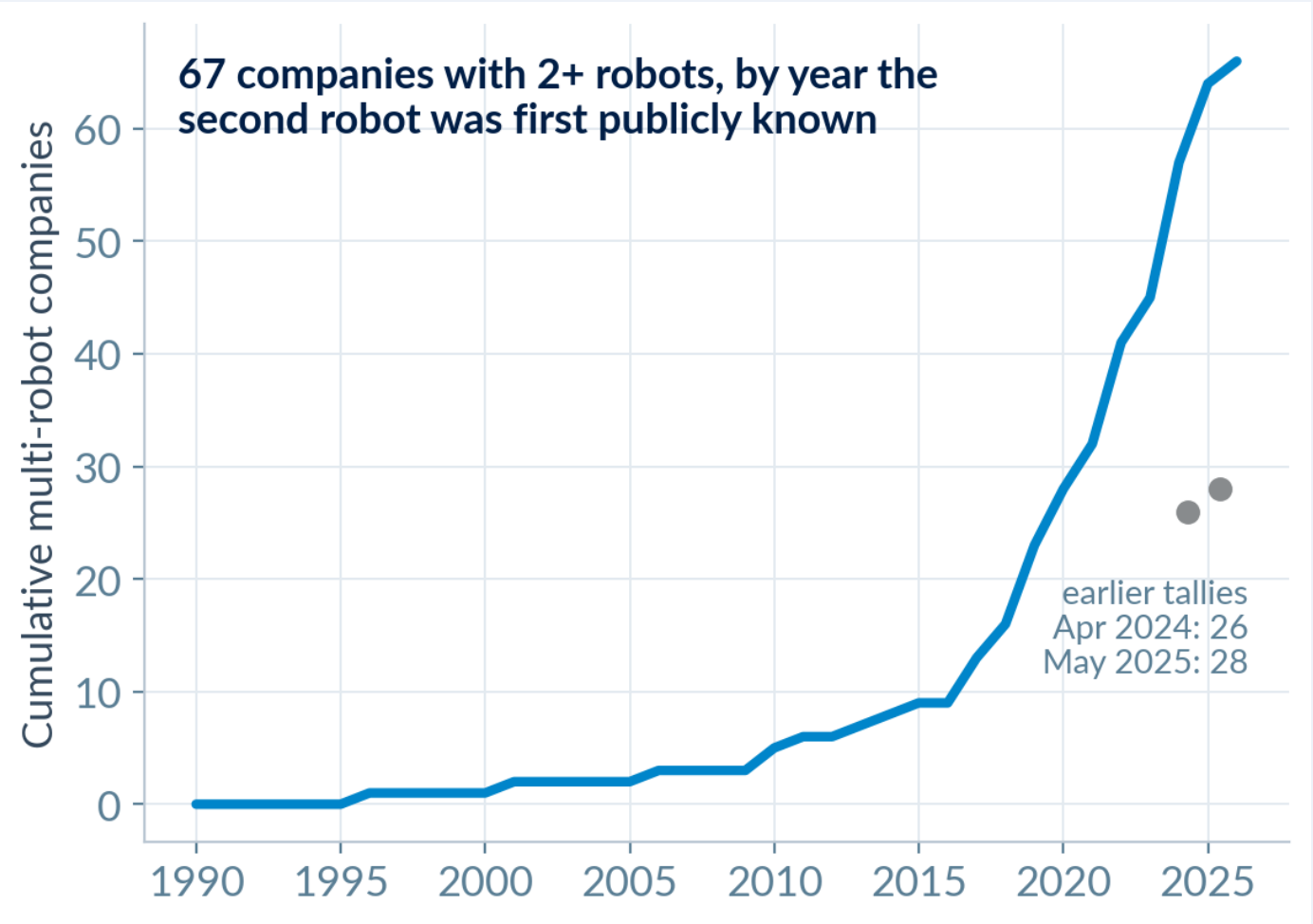


■ On Market (239) ■ Under Dev (371) ■ Acquired (46) ■ Dead (65) ■ Uncertain (39)

On Market re-verified against regulator databases & company disclosures. Rows are robots; companies counted once across multi-robot portfolios.

# Multi-robot portfolios are mostly recent

Roster · sources: patent filings, FDA/NMPA submissions, press releases, papers; collected 8-11 Jul 2026



## 67 MULTI-ROBOT COMPANIES

**10% of tracked companies field two or more robots**

Dated by first public evidence for each second robot: patent filing, FDA/NMPA submission, press release, or paper

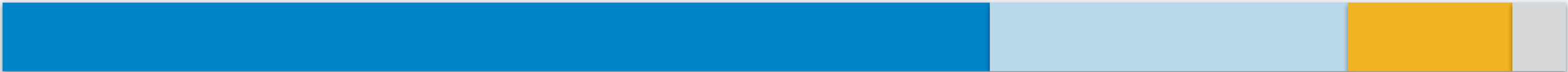
**57% went multi-robot in 2021 or later; earlier edition tallies (26, 28) lagged detection**

# How they work

Roster · control philosophy across all 760 robots; teleoperation still dominates



## CONTROL MODE OF THE 760 ROBOTS

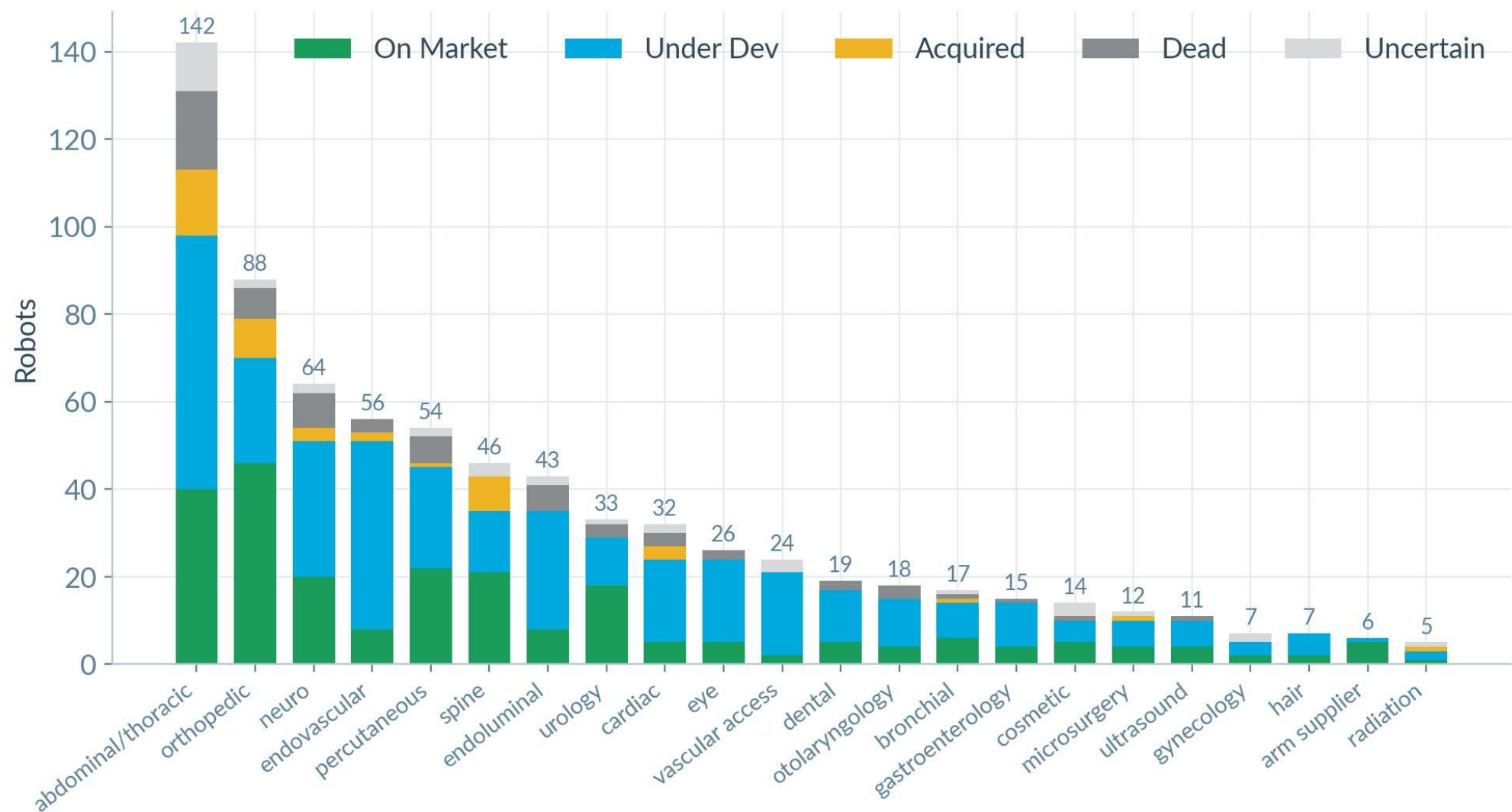


■ Teleoperation (480) ■ Preplanned (174) ■ Autonomous (80) ■ Handheld / manual / cooperative (26)

Autonomous roughly doubled since January (37→80), partly reclassification during re-verification, partly a real trend.

# Focus is still predominantly in abdominal procedures

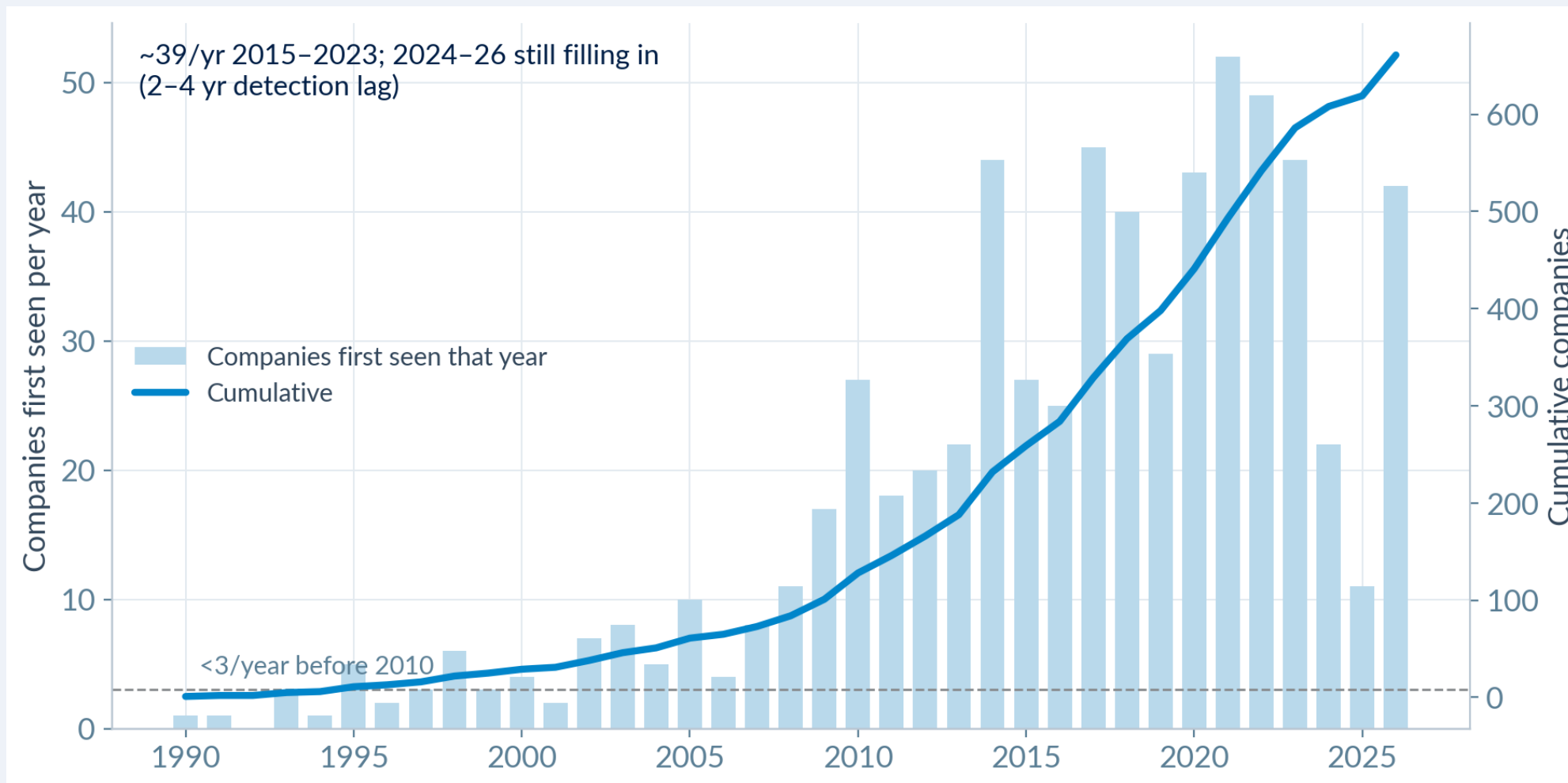
Roster · 760 robots by focus area, colored by status



**142 ABDOMINAL / THORACIC**  
**Orthopedic (88) and neuro (64) follow**  
A long tail of 32 focus areas  
**New categories since January: airway, regional anesthesia, trauma, podiatry, arm supplier**

# Company formation inflected around 2010

Growth · companies first seen per year, with the cumulative total



## 663 COMPANIES

**Fewer than 3 per year before 2010**

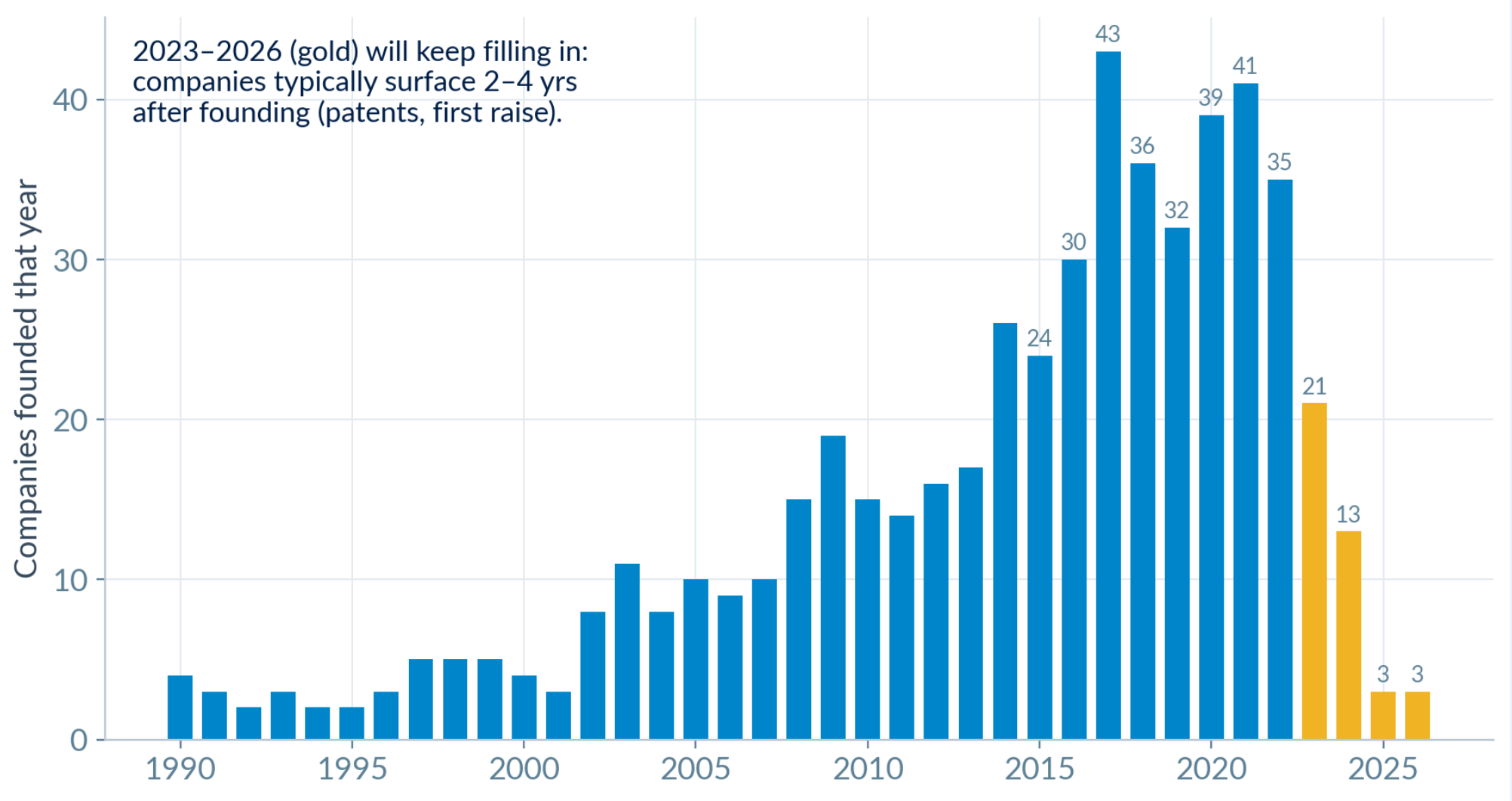
About 40 per year since 2015

**2024-26 look low only because companies take 2-4 years to become detectable**

**2026 shows a bump because of improved detection which affects first year of detection**

# By founding year, the field is younger than it looks

Growth · companies by founding year; the Founded column is populated for 566 of 663 companies



## DETECTION LAG

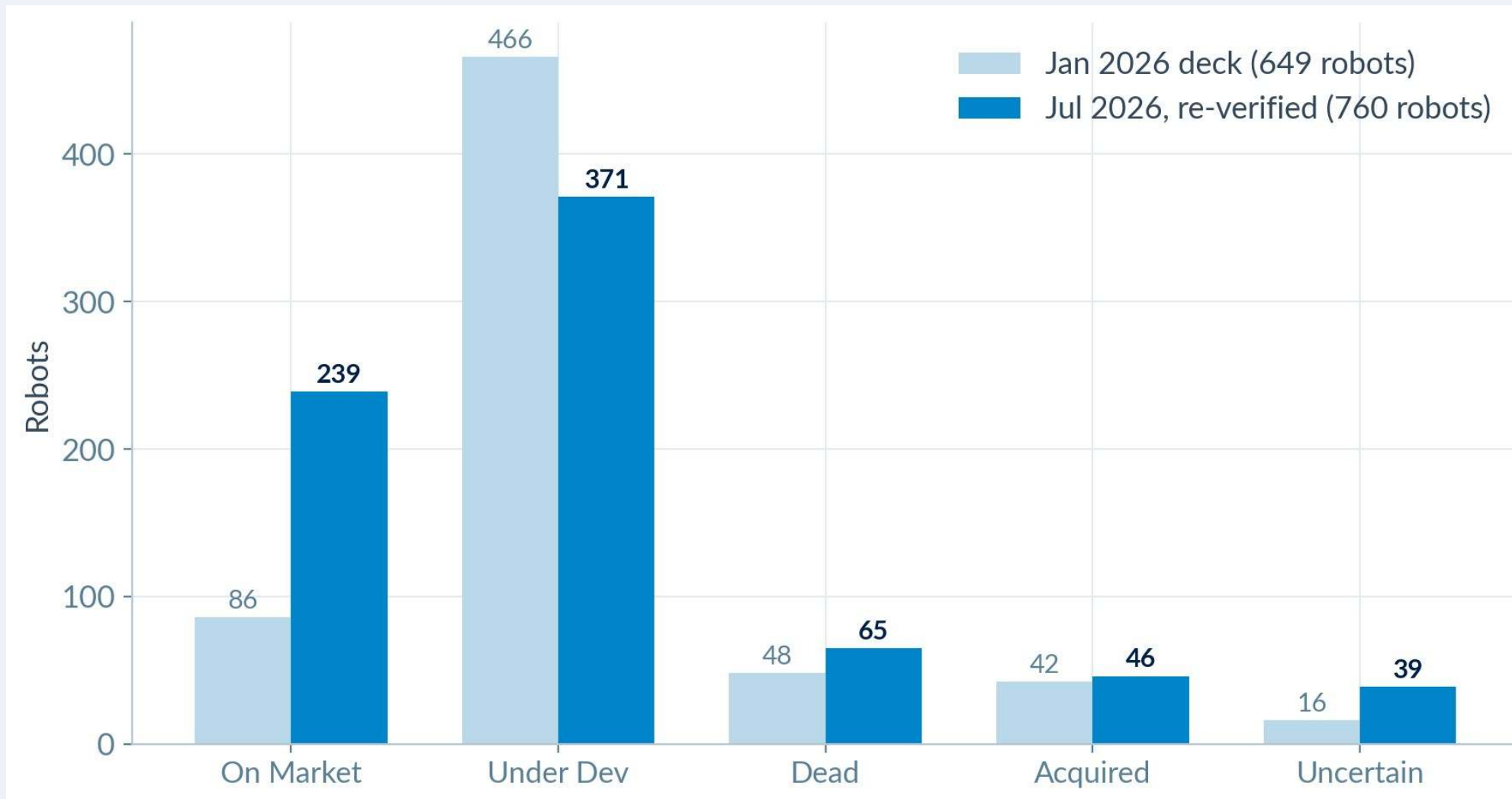
**Companies typically surface 2-4 years after founding**

First signals are patents and first raises, which lag incorporation

**Expect the 2023-2026 bars to keep growing in future editions**

# The commercial wave is bigger than it looked

Growth · re-verifying every company tripled the on-market count; China's approvals were undercounted



**86 → 239 ON MARKET**

## What changed

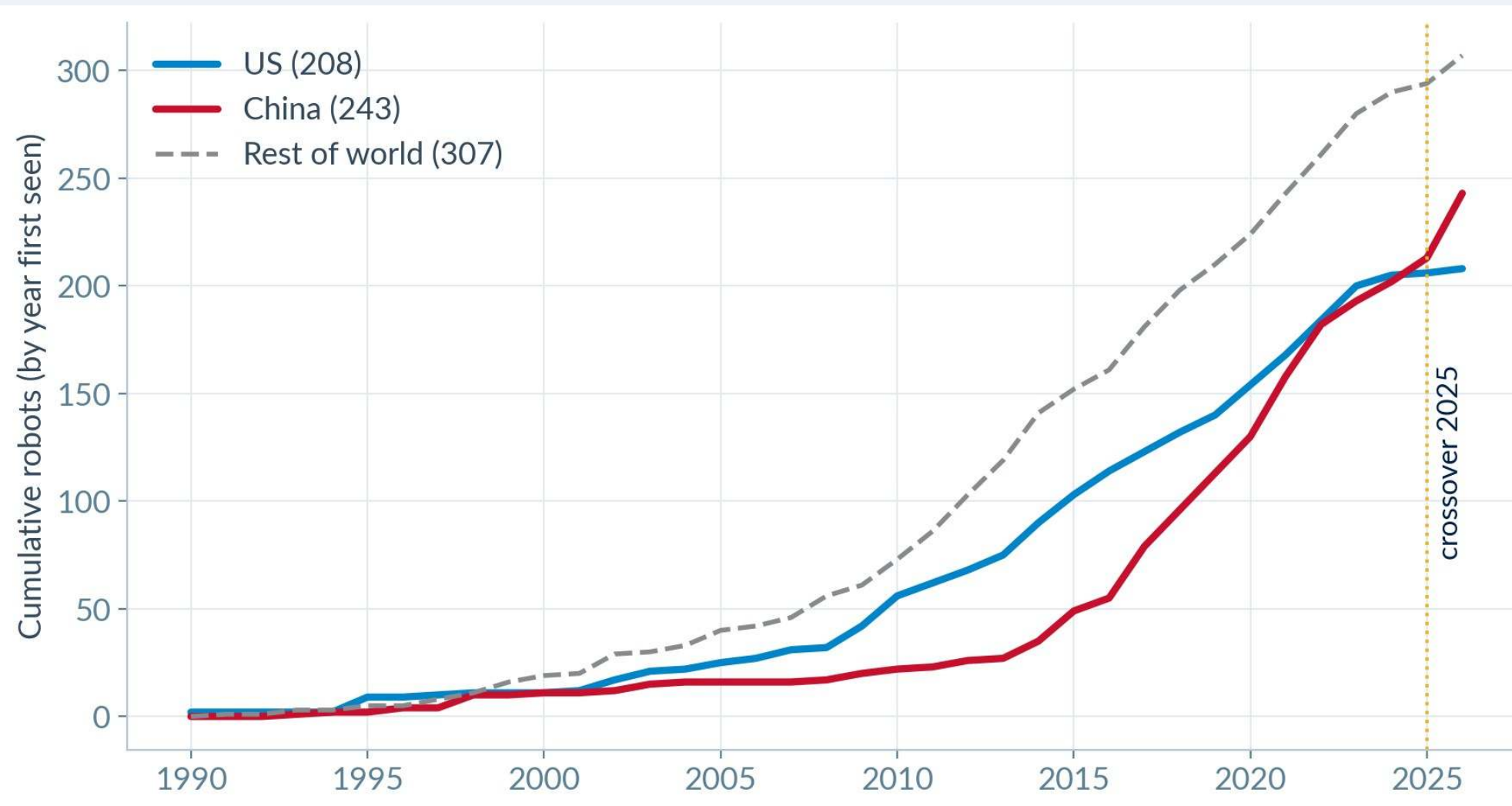
Jul 1-2 re-verification checked every row against NMPA/FDA/CE records and company disclosures

**Many Chinese systems had approvals & installs that never made English-language press**

Dead (48→65) and Uncertain (16→39) also grew, attrition was undercounted too

# China surpassed the US in total surgical robots

Geography · cumulative robots by year first seen, China 243 vs US 208 today



**243 vs 208**

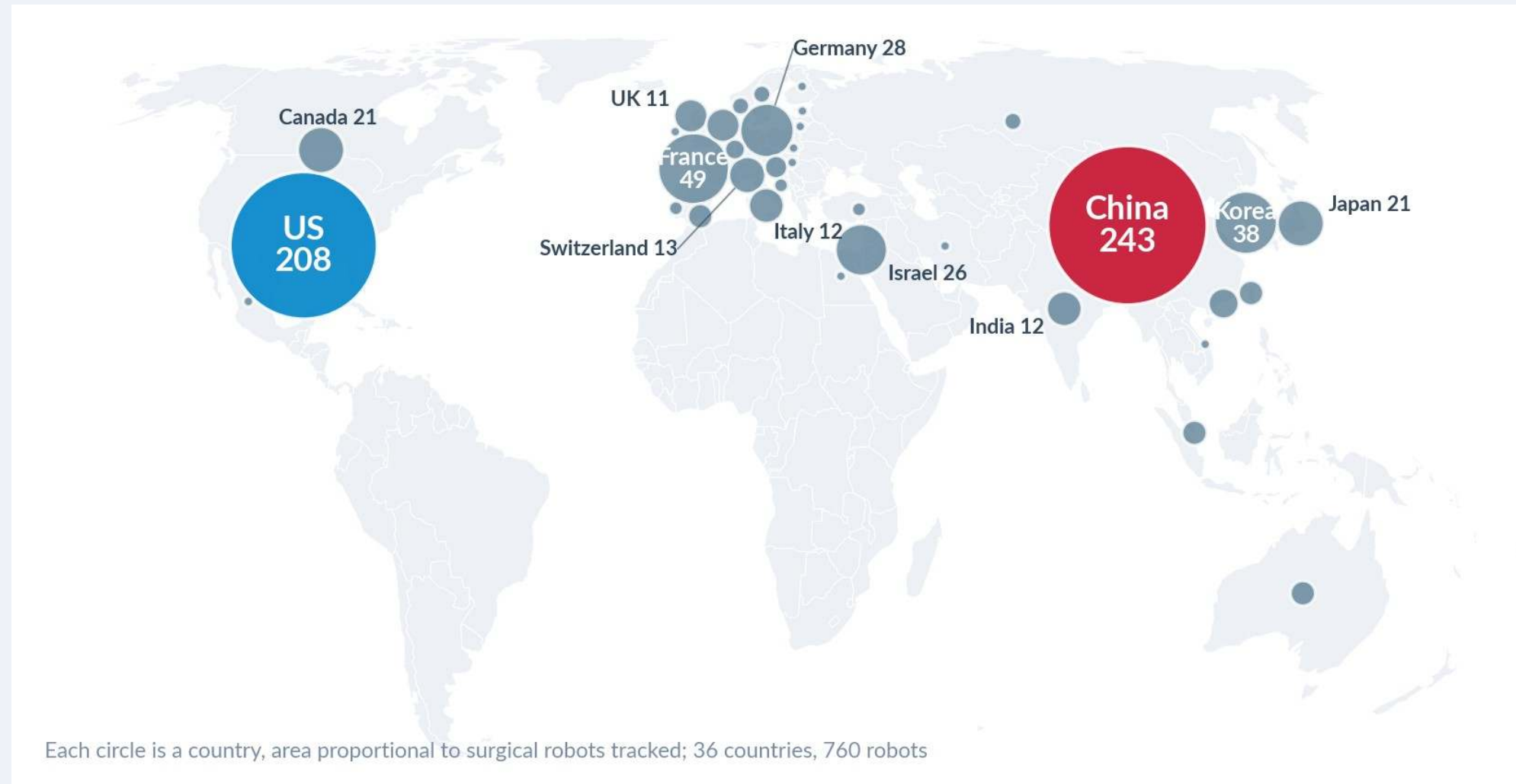
## China vs US robots

There is detection lag in the US/China counts

**Company-level counts mirror this crossover; detection lag on Chinese companies means the true gap is likely wider**

# Robots by country of origin

Geography · each circle is a country, area proportional to robots tracked



## 36 COUNTRIES

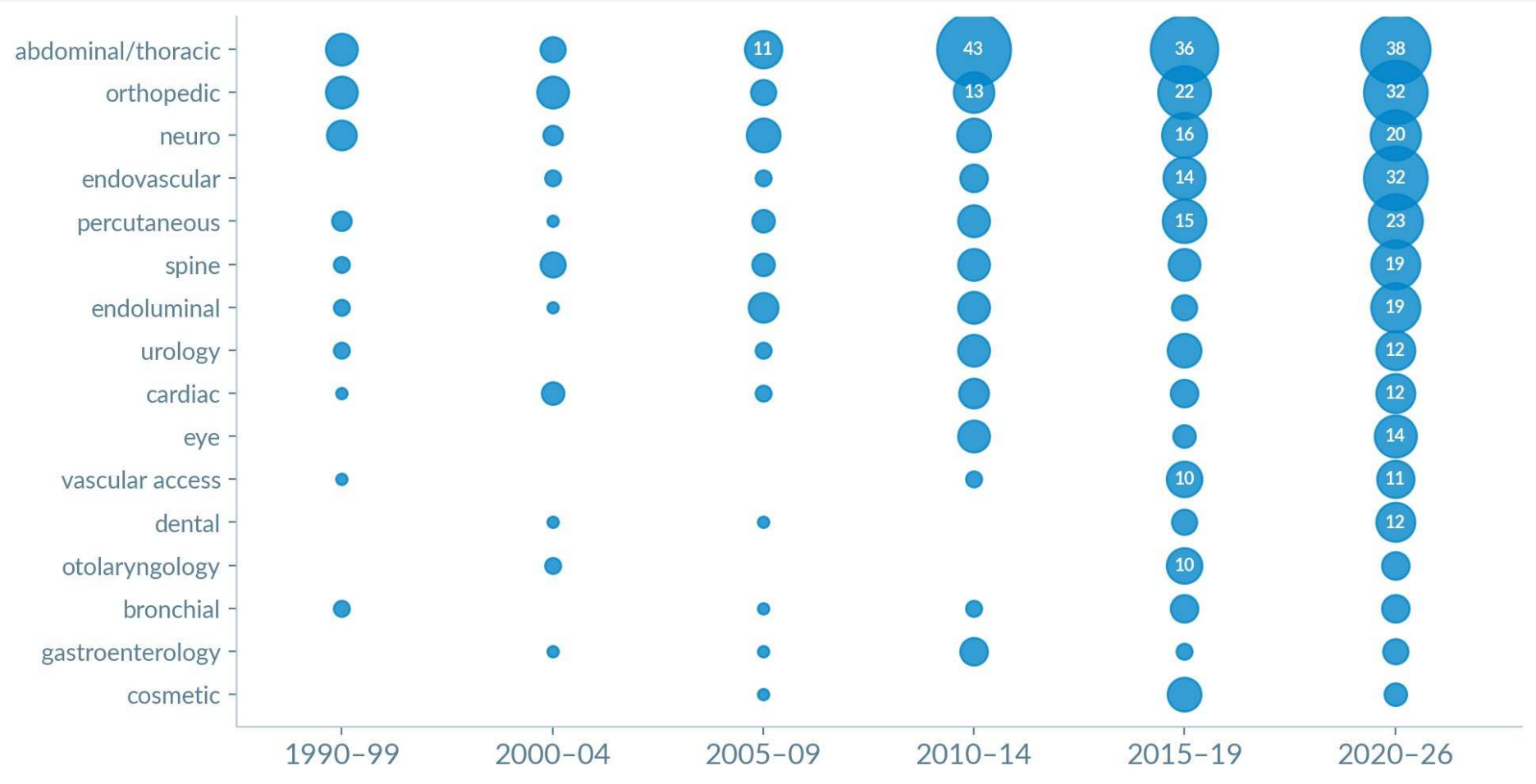
**US + China = 59% of all robots**

France (49), Korea (38), Germany (28) lead the tail

**China's share of new robots by era: 13% through 2014, 40% in 2015-19, 44% since 2020**

# Global specialty robots development start

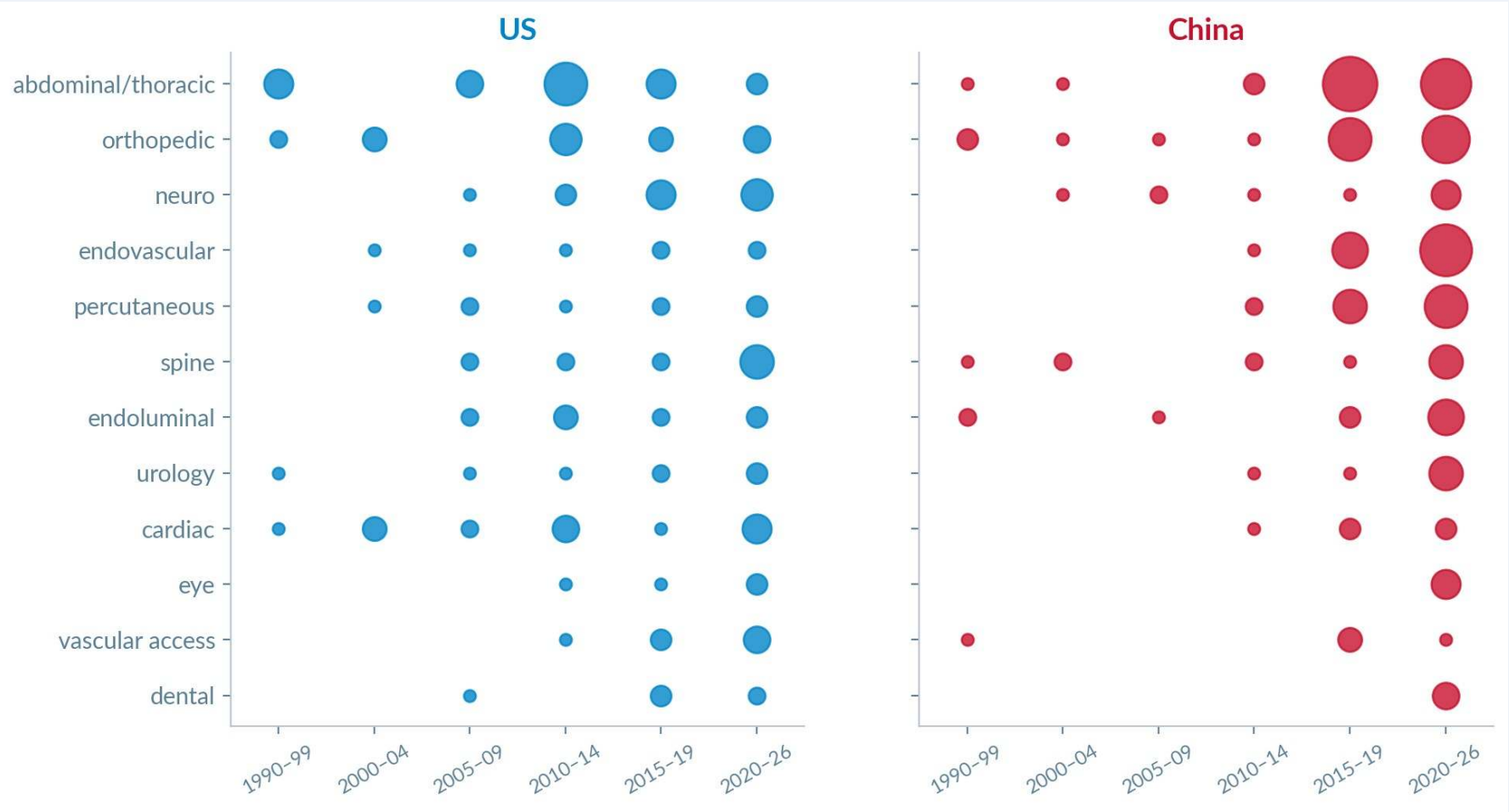
Specialty waves · Tufte-style period matrix; dot size = robots first seen in that period



Read across a row to see when each specialty became active; read down a column to see what each era produced.

# China and US robots begun by specialty

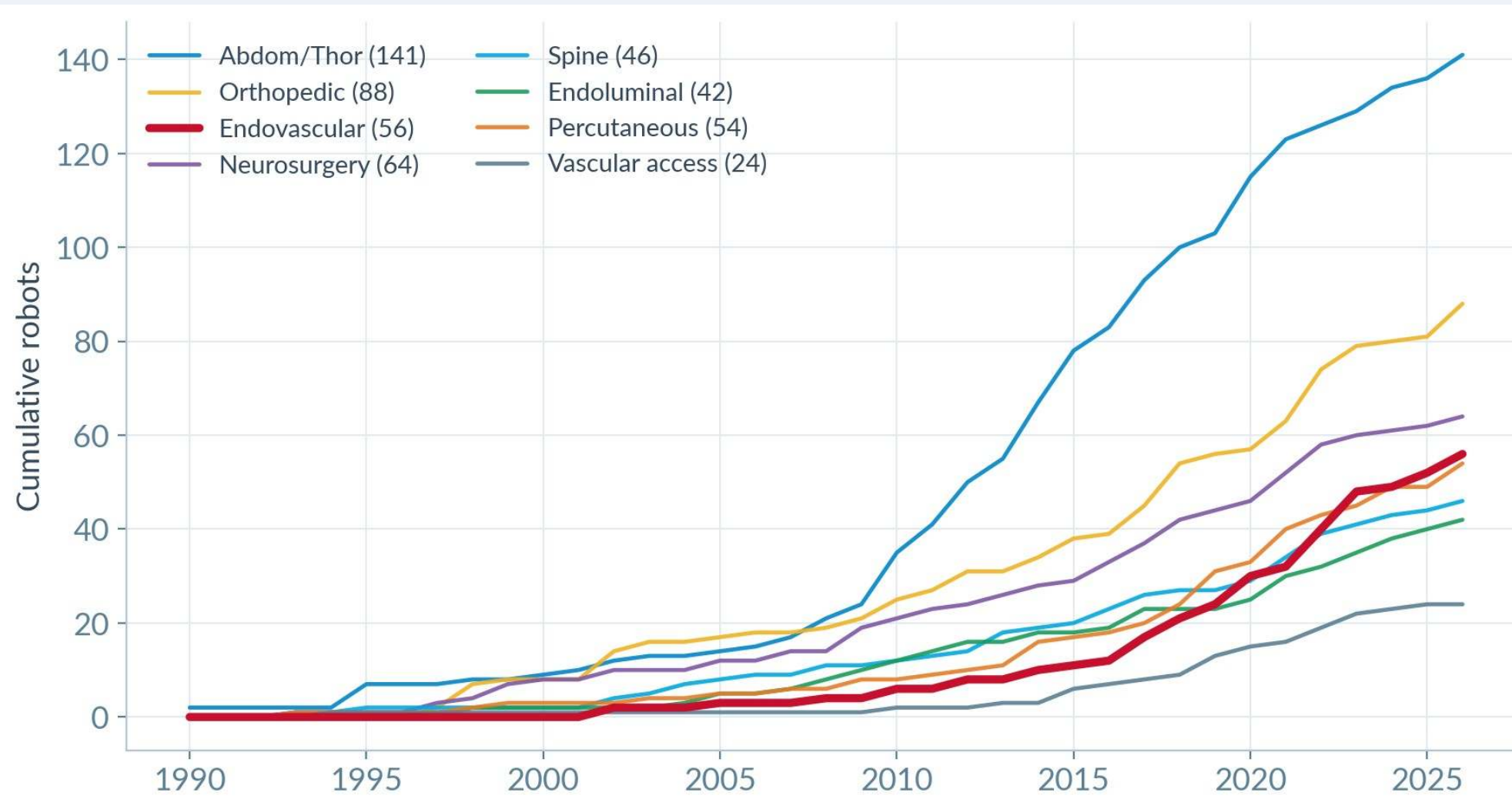
Specialty waves · side-by-side period matrices; dot size = robots first seen



The US spreads across specialties and decades; China concentrates hard in 2015-2026, especially abd/thor, endovascular, percutaneous and orthopedic.

# One specialty breaks the pattern

Endovascular case · cumulative robots by specialty; one curve does not fit the pattern



## 56 ENDOVASCULAR ROBOTS

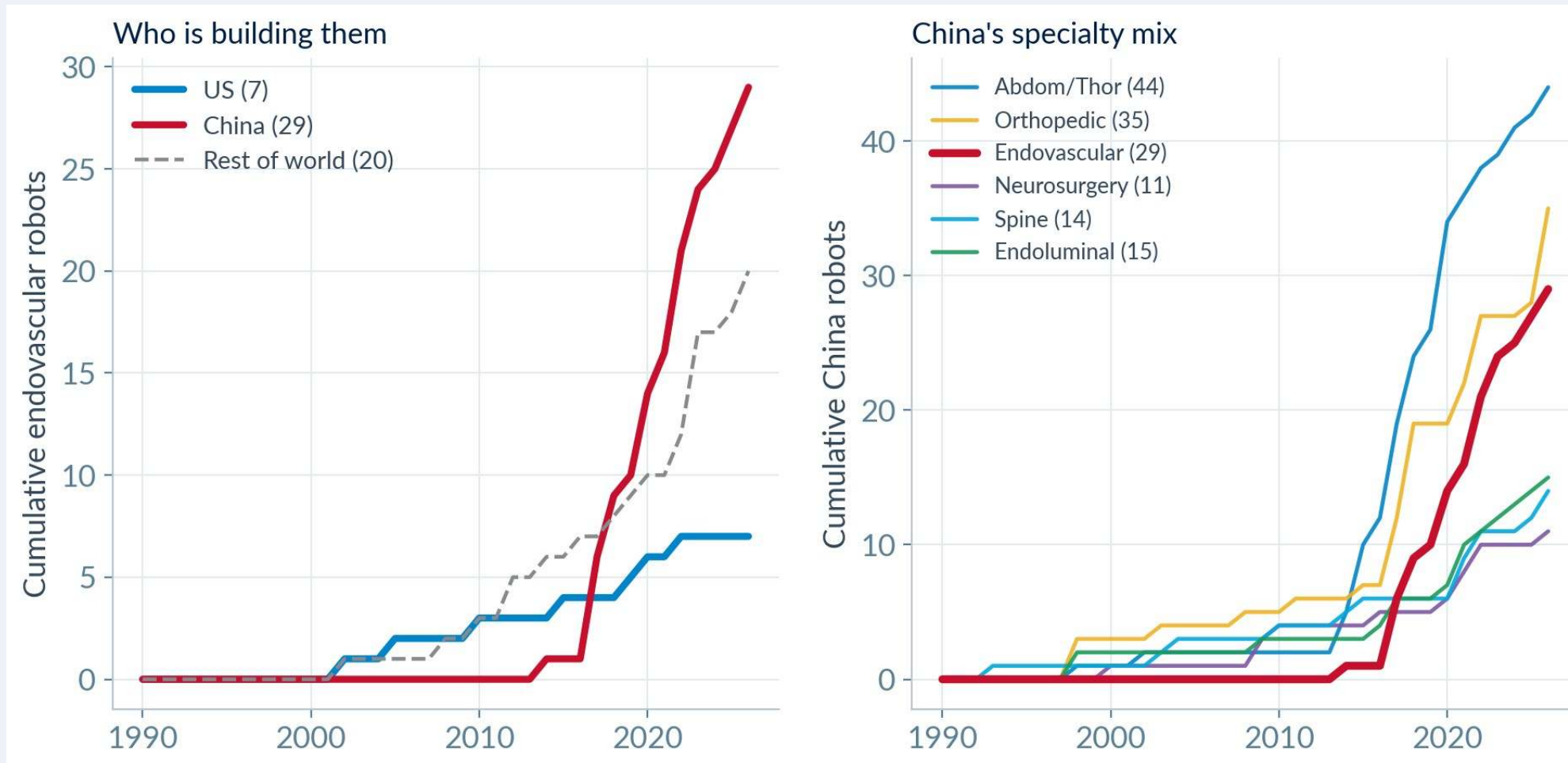
**Roughly 3/4 began after 2015**

The steepest specialty curve of the last decade

**China drives the spike; next slide**

# China accounts for the majority of the endovascular increase

Endovascular case · attribution in two views: who is building them, and China's specialty mix



## CHINA'S WAVE

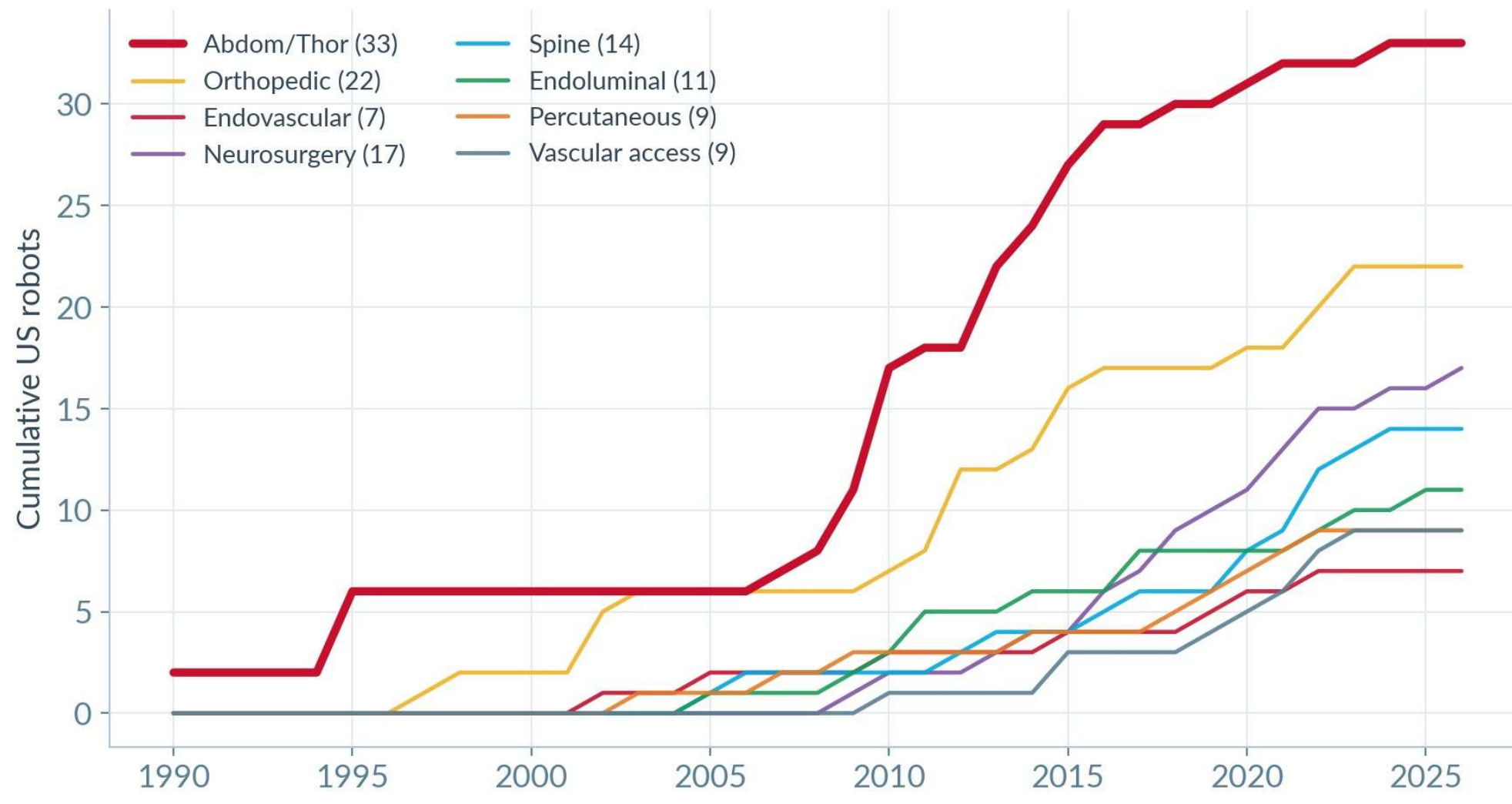
**Endovascular was a US and EU niche until about 2015**

China's curve bends sharply upward from 2018

**Within China, endovascular is among the steepest specialty curves**

# US: abdominal/thoracic wave, steady orthopedic growth

Endovascular case · the US contrast: cumulative US robots by specialty

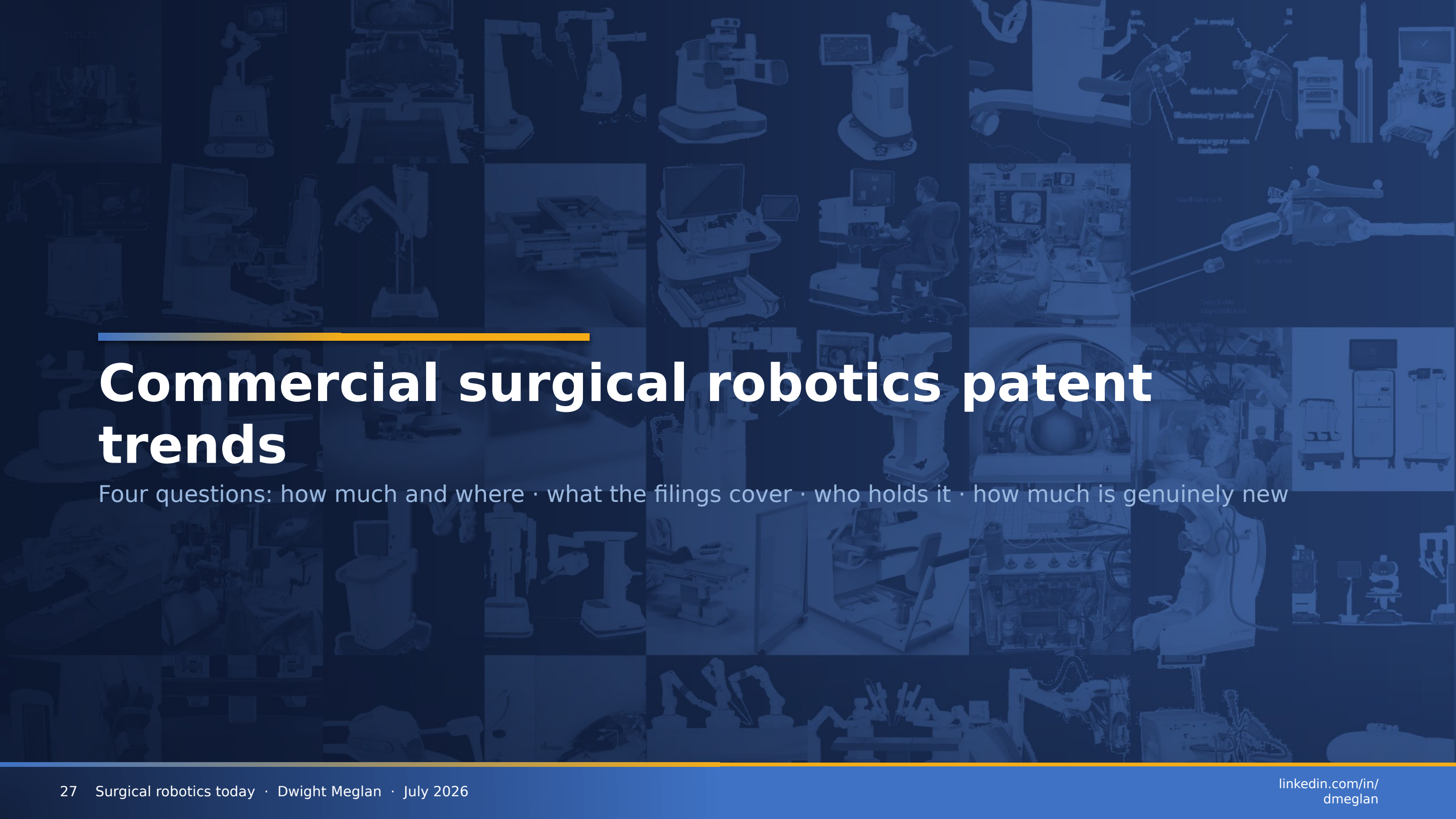


## US PATTERN

**Abdominal/thoracic keeps compounding despite the da Vinci incumbency**

Orthopedic growth has been steady

**Endovascular stays a modest US category, the contrast with China is the story**

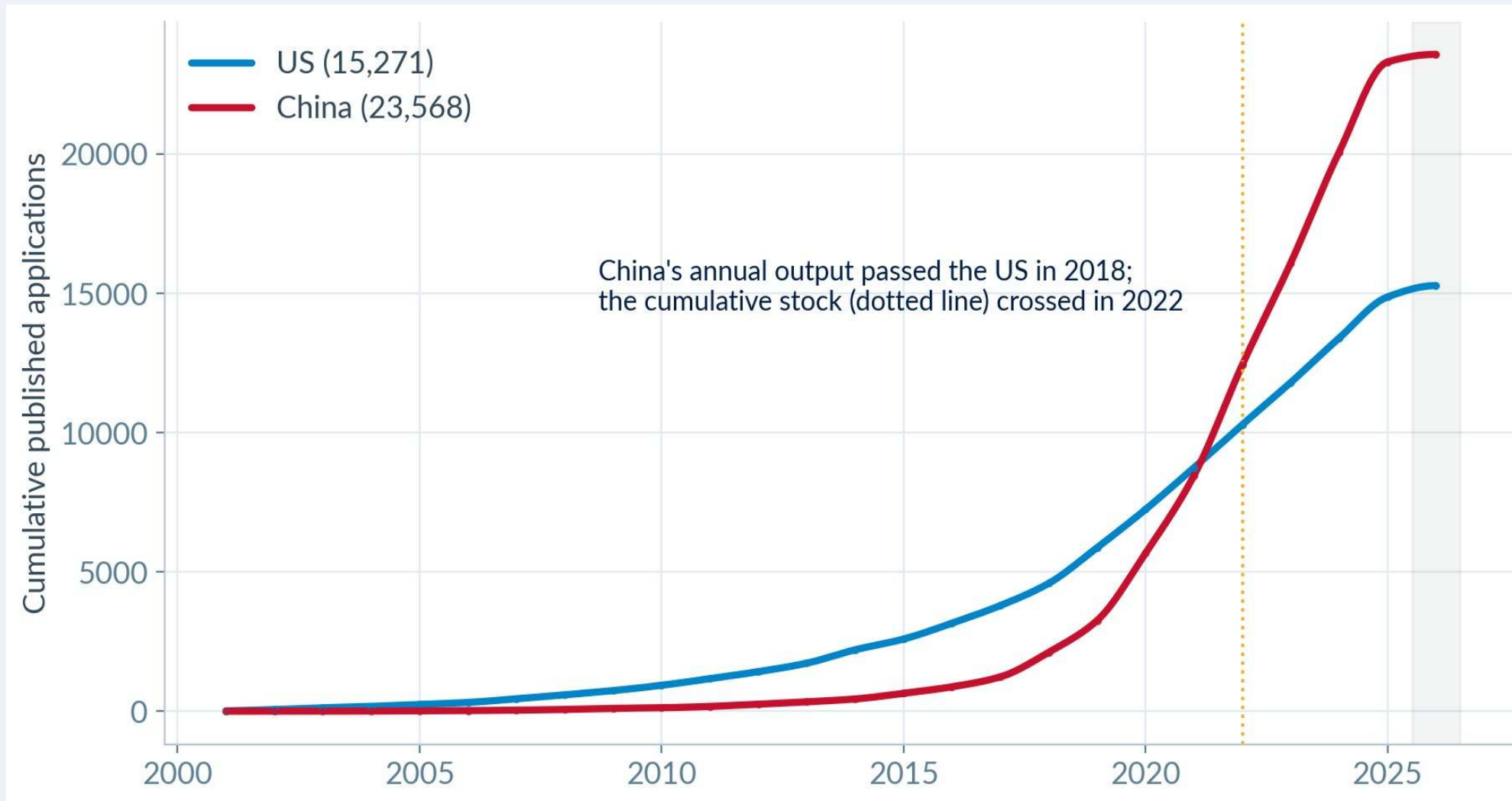


# Commercial surgical robotics patent trends

Four questions: how much and where · what the filings cover · who holds it · how much is genuinely new

# Overall surgical robotics patent applications: US & China

Cumulative published applications, A61B34/30 incl. subgroups, like-for-like basis, refreshed 7 Jul 2026



**23,568 vs 15,271**

## China vs US cumulative applications

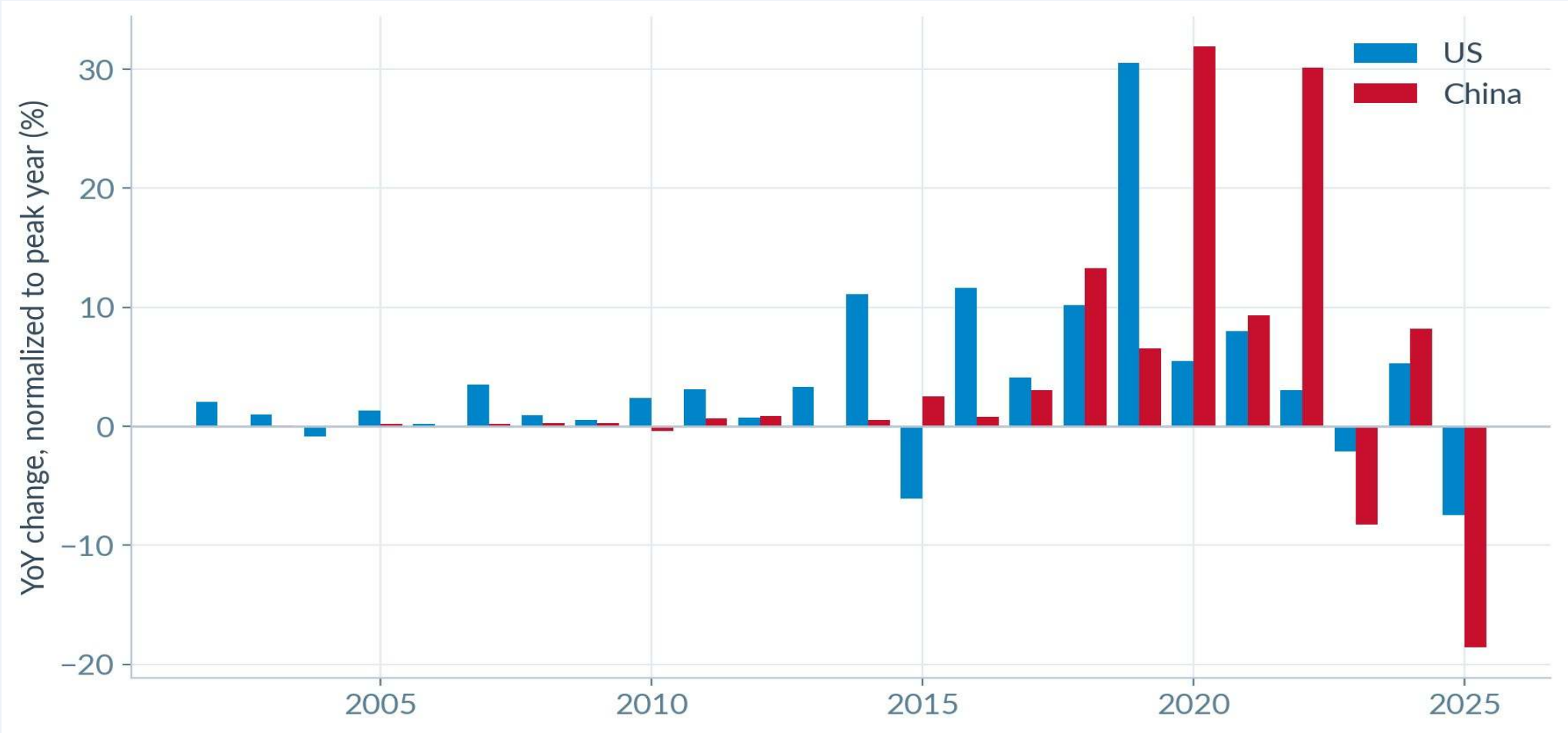
China's annual output passed the US in 2018; the cumulative stock crossed in 2022

**Earlier editions mixed counting bases and overstated the gap; like-for-like it is ~2.5x and widening**

**Source: Google Patents result counts, q=(A61B34/30) country:US/CN, applications only, by publication year; collected 7 Jul 2026**

# US & China patents: normalized year-on-year change

$(\text{Year}[n] - \text{Year}[n-1]) / \text{peak year, normalized to each country's maximum}$



## CHINA'S MOMENTUM

**China's filing rate grew faster than the US rate in most years since 2010**

Both countries decelerate after their peaks: China 2022, US 2024

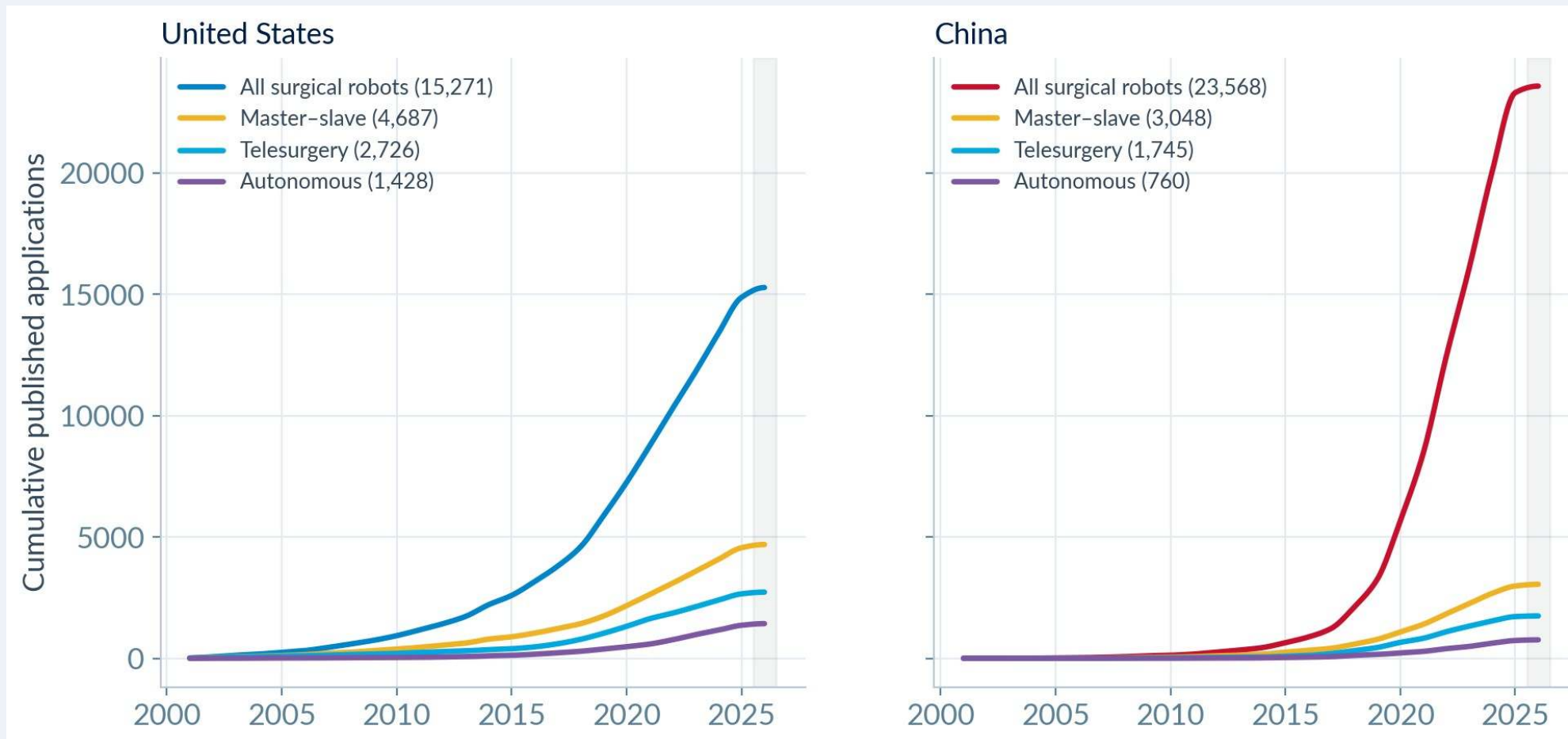
**2023-2025 values are understated by the ~18-month publication lag; treat recent bars as provisional**

**Source: Google Patents counts, same query as the previous slide**

Normalization anchors: US peak 1,595 published applications (2024); China peak 3,980 (2022). 2026 excluded (partial year).

# What the filings cover: US and China by CPC subgroup

Cumulative published applications by CPC subgroup; both panels share the same vertical scale. Source: Google Patents result counts,  $q=(A61B34/3x)$ , applications only, by publication year; collected 7 Jul 2026



## SAME CATEGORIES, DIFFERENT SCALE

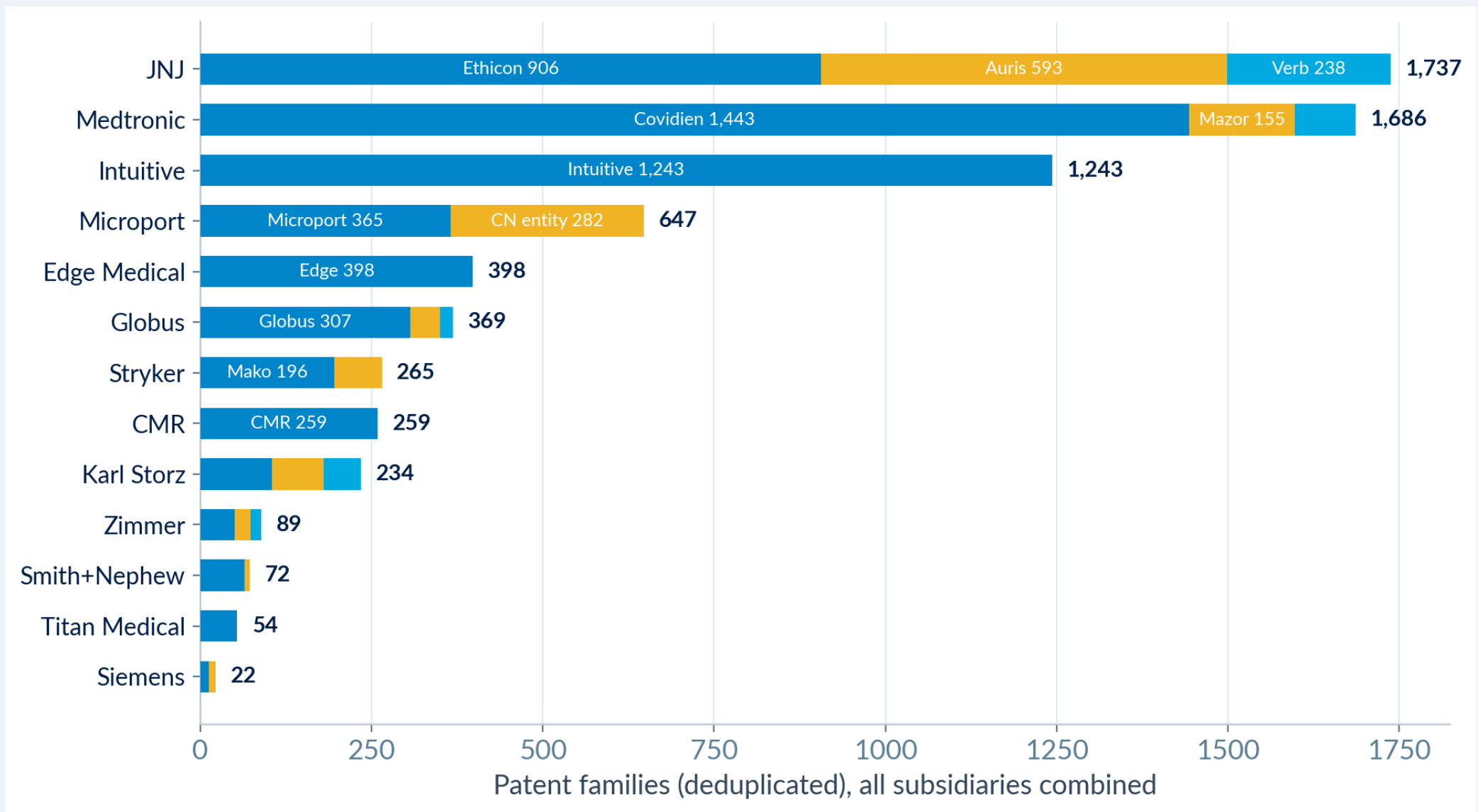
**All surgical robots (34/30 incl. subgroups), master-slave, telesurgery, autonomous**

China's curves accelerate the most after 2018

**Source: Google Patents result counts,  $q=(A61B34/3x)$  with country and status filters, by publication year; collected 7 Jul 2026; 2026 is first-half only**

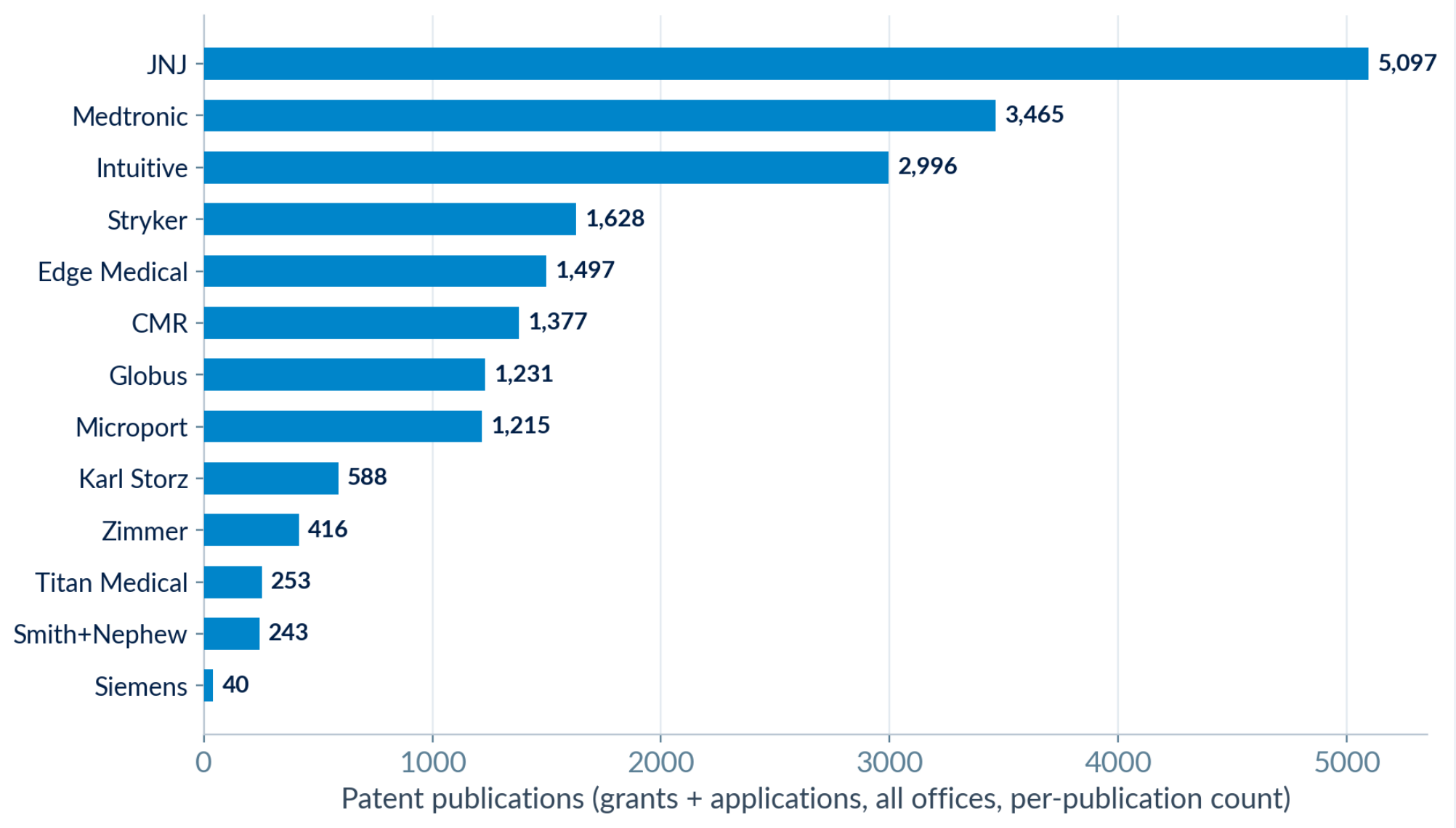
# For IP, you need to count all the parts

Patent families by holder, subsidiaries stacked separately. Source: Google Patents result counts, TI/AB/CL=(surg\* NEAR/3 robot\*) per assignee, family-deduplicated; collected 12 Jul 2026



# While there have been many patents published...

Publications (grants + applications, all offices) for the largest holders. Source: Google Patents result counts, per-publication mode; collected 12 Jul 2026



## RAW VOLUME

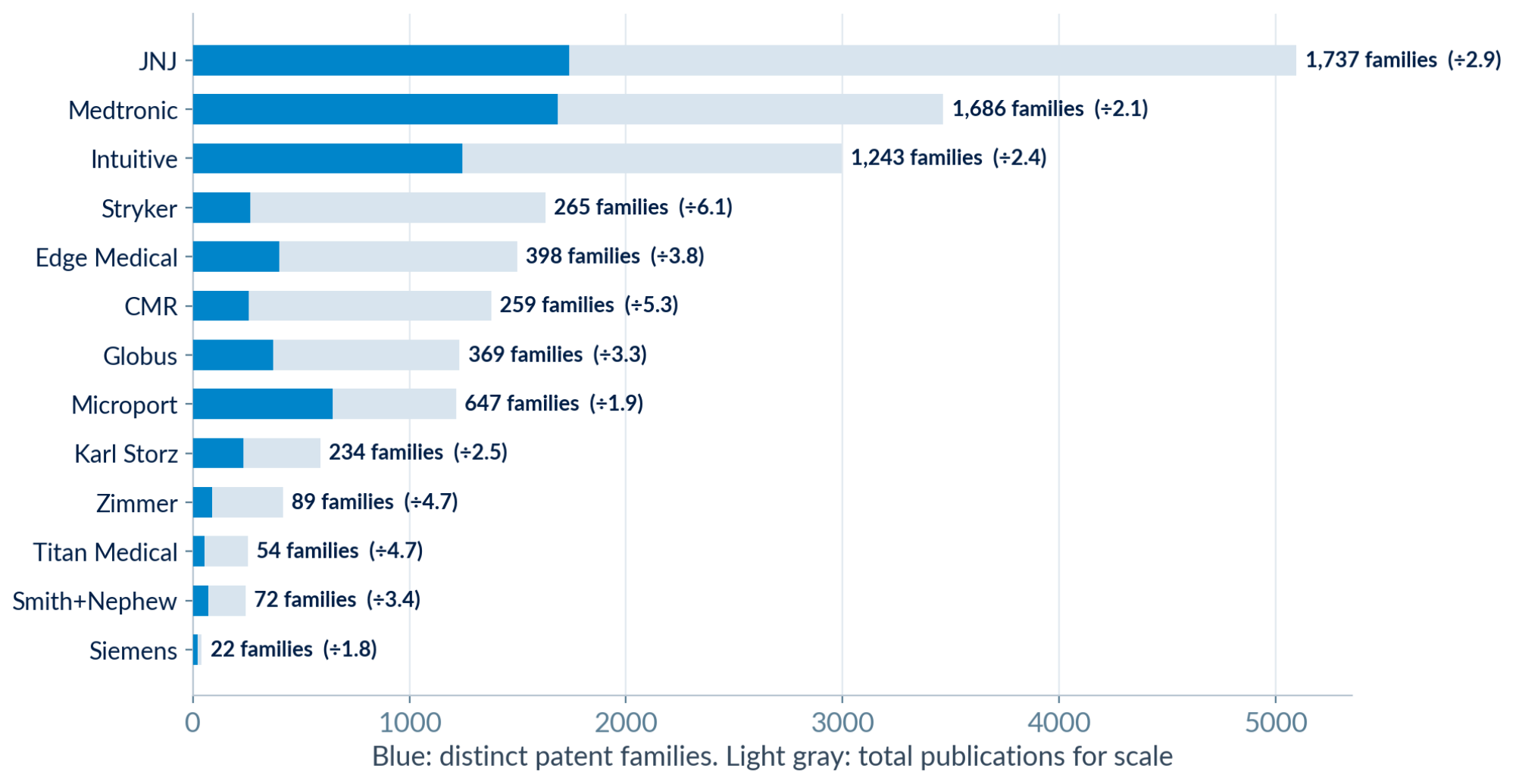
**JNJ group (Ethicon + Auris + Verb): 5,097 publications; Intuitive alone: 2,996**

Publication counts include every office and every continuation

**Volume alone overstates novelty, next slide**  
**Source: Google Patents, collected 12 Jul 2026**

# ...many are extensions of existing IP

Distinct patent families vs total publications, same query. Source: Google Patents result counts, family-deduplicated vs per-publication; collected 12 Jul 2026



## FAMILY-LEVEL VIEW

**Families run 2.5x to 6x below publication counts (Stryker 6.1x, CMR 5.3x, Intuitive 2.4x)**

JNJ group: 1,737 families across Ethicon, Auris, and Verb

**For diligence, count families and claims, not publications**

**Source: Google Patents, collected 12 Jul 2026**



# Examples of surgical robots in different specialties

Pictures shown not remotely comprehensive: intentionally chosen to provide historical perspective  
Note that DEAD classification refers to the robot, not necessarily the company

# Abdominal / Thoracic

Representative examples



# Abdominal / Thoracic

142 robots from 127 companies

34

## ON MARKET

Agibot Medtech  
AKTORMed  
Applied Dexterity  
Asahi Surgical Robo...  
CMR Surgical  
Cornerstone Robotics  
Dextérité Surgical  
Distalmotion  
Edge Medical Roboti...  
FreeHand Surgical  
Genesis MedTech Gro...  
HIWIN Technologies ...  
Huger Medical Robot...  
HumanTouch Surgical  
Innolcon / 以诺康  
Intuitive Surgical  
JNJ MedTech  
Levita Magnetix  
LivsMed  
Meere Company  
Merai Newage  
Moon Surgical  
Parallel Robotics L...  
Precision Robotics  
Ronovo Surgical / 瑞...  
Sagebot / 思哲睿

Sina Robotics and M...  
SS Innovations  
Tuebingen Scientifi...  
Tuodao Medical  
United Imaging Surg...  
Virtual Incision Co...  
Weigao Group / 威高集团  
Wise King / 唯精医疗

51

## UNDER DEV

Accrea Medical Robo...  
All-Star Medical  
Aopeng Medical / Op...  
Apeiron Surgical / ...  
Articulus Surgical  
Asvel OÜ  
B Braun New Venture...  
Baining Yingchuang ...  
Borns Medical Robot...  
Canady Robotics  
DEKA Research & Dev...  
Denovo Bioinnovatio...  
Doosan Robotics  
Eabmed Science and ...  
Erbe Elektromedizin...  
Fulbright Medical  
Gamania Digital Ent...  
IMRobonix  
Jinshan Science & T...  
Kirubotics Surgical...  
LapTics Surgical  
Macheon AI  
Maituo Medical  
Mars Medical Robot  
Meike Electronic Eq...  
Micron Shiga / 株式会社...

MindAdvance Medical...  
Mindray / 迈瑞医疗  
Nikkotech  
NISI  
Noahtron Intelligen...  
Nuclear Medical Rob...  
Olympus Corporation...  
Optosurgical, LLC  
Panther Healthcare ...  
Rainbow Robotics / ...  
Revolve Surgical  
Rob Surgical System...  
Ruida Medical Devic...  
Scichip Robotics  
Shengzhi Medical / ...  
Shukra Robotics Pvt...  
Steerable Instrumen...  
Symphera GmbH  
Titan Medical / Con...  
ValueBioTech S.R.L.  
Vicente Medical Rob...  
Viestar  
VinSurgical  
XSurgical Robotics  
Yinuoda Medical / 易...

14

## ACQUIRED

Asensus Surgical  
Auris Health / JNJ  
Computer Motion  
Covidien  
Endocontrol SA  
Gomtec / ABB  
Great Belief Intern...  
Human Xtensions  
JNJ Ethicon Endo-Su...  
Medical Surgery Tec...  
SOFAR S.p.A.  
TransEnterix  
Tuge Healthcare  
Verb Surgical

11

## UNCERTAIN

Assisted Surgical T...  
B Braun  
Ekymed  
EndoRobotics, Inc.  
Kexhixing Robot  
Medsys SA  
Modular Surgical  
Movasu  
Shuchuang Robotics  
SRI International  
Virtu Surgical

17

## DEAD

Agile EndoSurgery  
American GNC Corpor...  
Avatera Medical  
AVRA Medical Roboti...  
Bio-Medical Enginee...  
Human Robotics S.A....  
Invivo Robotics  
Lapara Surgical  
LG Electronics  
MDA  
Omniboros  
Panasonic Holdings ...  
Roboscopy Medical L...  
Samsung Electronics  
Signature Robot Ltd  
SPI Surgical, Inc.  
Vicarious Surgical ...

# Orthopedics

Representative examples



# Orthopedics

88 robots from 77 companies

37

## ON MARKET

Aimooe / 艾目易  
Airs Inc  
AK Medical  
Amplitude Surgical  
Chunli / 春立医疗  
Corin Apollo / ApolloKnee  
Curexo / 큐렉소  
DePuy Synthes  
DroidSurg Medical  
Enovis  
FUTURTEC Medical  
Globus Medical  
Gusheng Yuanhua Robot  
Howmedica Osteonics Corp.  
Hurwa / 和华瑞博  
Jianjia Medical Technolo...  
Kuangrui Intelligent Tec...  
Lancet Robotics  
Longwood Valley MedTech  
McGinley Orthopedics  
Medos  
MedTech SA  
Merai Newage  
Meril Healthcare  
MicroPort MedBot  
Naton Medical Robotics

Rossum Robot Technology  
Santa Medical Technology  
Smith & Nephew plc  
Stryker  
Surgify Medical  
THINK Surgical / Curexo  
Tinavi Medical Technolog...  
Tuodao Medical  
Unik Orthopedics, Inc.  
Weigao Group / 威高集团  
Yidian Lingdong Technolo...

22

## UNDER DEV

Aopeng Medical / Operati...  
ArthroLase  
Bozhi Bio  
Connecteve  
Convergence Medical  
Dading Medical Technolog...  
Duluth Medical Technolog...  
Extremis Robotics  
Ganymed Robotics  
Great Robotics / 歌锐科技  
Hangwei  
Interventional Systems  
Jieduo  
Jimuu Intelligent Tech C...  
Kyniska Robotics  
Laishbo Medical Technolo...  
Linkwah Technologies, In...  
Medical Robotics i Stock...  
Robossis, Inc.  
Tianxing Medical Co.  
Xingyun Guiyi Intelligen...  
Xinkangyuan Minimally In...

9

## ACQUIRED

Blue Belt Technologies  
KAT Robotics  
LinkX Robotics  
Mako Surgical Corp.  
Monogram Technologies  
OMNIlife Science  
OrthoSpin Ltd.  
Orthotaxy SAS  
Ostesys

2

## UNCERTAIN

Seva Robotics  
Shukang Medical Technolo...

7

## DEAD

Acrobot / Stanmore  
Advanced Osteotomy Tools  
iin medical  
Longhui Medical / 上海龙慧医  
疗  
ortoMaquet GmbH / Maquet  
Revision Robotics Surgic...  
Sagitta Robotics Limited

# Spine

Representative examples



# Spine

46 robots from 45 companies

20

## ON MARKET

ATEC  
Bangjiexing Medical Technolo...  
Curexo / 큐렉소  
Cyber Surgery ALAYA  
Dabo Medical / 大博医疗  
eCential Robotics  
LEM Surgical  
MedTech SA  
Medtronic  
Meyer Optoelectronic Technol...  
MinMaxMedical  
Perlove Medical / 普爱医疗  
Point Robotics MedTech / 炳碩生醫  
Remex Medical Corp.  
Sino-Europe MicroRobotics  
SMAIO SA  
SMTP Technology Co., Ltd.  
Sylorus Robotics  
Xinjunte Smart Medical Equip...  
Zhuzheng Robot / 铸正机器人

14

## UNDER DEV

8i Robotics  
Carevature Robotics  
Chunfeng Huayu Smart Medical  
Deneb Medical  
Dianhe Medical Technology  
FoxEyes Corporation  
Kato Medical / KatoMed  
Koellner Medical Technologies  
Linyan Medical / 上海霖晏医疗科技  
Maestro Surgical Robotics Ltd.  
Robotron Surgical Technologies  
Sanyou Medical  
Surgitec Robotics  
Weigao Group / 威高集团

8

## ACQUIRED

Cardan Robotics  
Excelsius Surgical  
Fusion Robotics / Accelus  
KB Medical  
Mazor Robotics  
Mobius Imaging  
Mobius Imaging / Stryker  
Relievent Medsystems

3

## UNCERTAIN

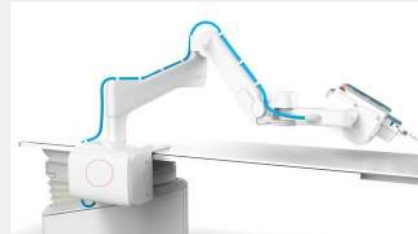
MI4Spine, LLC  
Rubicon Spine, LLC  
Teslake, Inc.

# Endovascular

Representative examples



**Corindus**  
A Siemens Healthineers Company



**Robocath**

*Catheter Robotics*



**microbot**  
medical



**hansen**  
MEDICAL



**STEREOTAXIS**

**aeon scientific**



# Endovascular

56 robots from 55 companies

8

## ON MARKET

Aibo Medical Robot  
Artiria Medical  
Inovatyvi Medicina  
Microbot Medical  
MicroPort MedBot  
UnionStrong  
WeMed / 唯迈医疗  
Yidu Hebei Robot Technology

42

## UNDER DEV

Angio8 / Shenzhen Tongyong T...  
Aopeng Medical / Operation R...  
Apeiron Surgical / 极限人工智能  
Basecamp Vascular  
Biomedical Research Center LLC  
Biotronik AG  
Calabrian High Tech  
Curie Interventional Robotics  
Dashì Technology  
Dinova Neuroscience Technolo...  
EndoWays / Ascendum  
Flux Robotics  
Great Robotics / 歌锐科技  
Hanglok-Tech  
Huada Zhizao Yunying Medical...  
Huaxi Jingchuang Medical Tec...  
InterMag  
Jieruo Medical Technology  
Jingge Medical Technology Co.  
Koswire / 코스와이어  
Langbokang Medical Technology  
Langya Hechuang Medical Tech...  
Magellan Biomedical Inc.  
MagnebotiX  
Nano4Imaging  
New Vascular

Nimble Surgical  
Perazah Inc. / 페라자  
RainMed Medical  
Raysight Medical  
Remedy Robotics  
ROBGENIX MEDICAL PTE. LTD.  
Route 92 Medical  
Shenganjie Medical Technology  
Shengyi Technology  
Terumo Corporation  
Vital Biomedical Technologies  
White Bear Medical LLC  
Wirescience Medical Technology  
Yindatong / 盈达通  
Zero Radiation Intervention  
Zhongke Hongtai Medical Tech...

2

## ACQUIRED

Corindus  
RoboCath

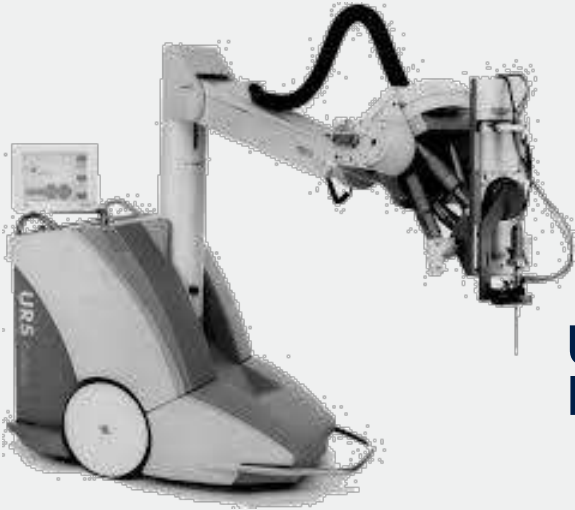
3

## DEAD

Aeon Scientific  
Catheter Precision, Inc.  
Hansen Medical / JNJ

# Neurosurgery

Representative examples



Universal  
Robot Systems



# Neurosurgery

64 robots from 62 companies

19

ON MARKET

ANT Neuro  
Ariemedi / 艾瑞迈迪  
Axilum Robotics  
B Braun  
Brain Navi Biotechnology  
Brainlab  
Elekta AB  
HOZ Medical / 华志微创  
IMRIS Imaging  
Insightec  
Koh Young Technology / 고영테크놀로지  
MedTech SA  
NovaSignal / Neural Analysis  
Rapid Medical  
Remebot / 柏惠维康  
Renishaw plc  
Sinovation Medical  
Vas Medical  
Zeta Surgical

30

UNDER DEV

AiM Medical Robotics  
Bionaut Labs  
ClearPoint Neuro  
Epica International  
Esspen GmbH  
Ezmedibot  
Genlight MedTech  
HROCT LITT Medtech / 超光微  
...  
IMSystem / 아임시스템  
Inetec Nero  
Magnendo  
Nanoflex Robotics  
Navigantis  
Neuralink  
neuro42  
Neurosputnik  
NeuroXess  
OrbSurgical Ltd.  
Panda Surgical  
Project 1985, LLC  
Robeauté  
Rogue Research  
RONNA Medical  
SceneRay / 景昱医疗  
Sibo Huiyi Tech / 思博慧医科技  
Siemens Healthineers End...

Telos Health  
UNandUP  
Weigao Group / 威高集团  
XCath

3

ACQUIRED

ArteDrone  
MDA  
Medineering / Brainlab

2

UNCERTAIN

Signate Life Sciences  
TheraSonic

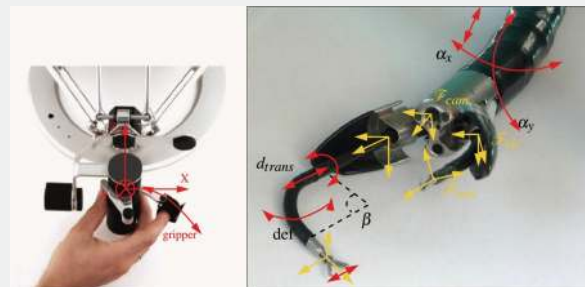
8

DEAD

Centauri Robotic Surgica...  
CranioMedix Inc.  
Euphrates Vascular  
ISIS  
MDA Space Ltd.  
NeuroArm / SYMBIS  
Siemens Healthineers End...  
Synaptive Medical

# GI Endoluminal

Representative examples



**STORZ**  
KARL STORZ — ENDOSKOPE



endoVia



**ENDOQUEST**  
ROBOTICS



# Endoluminal

43 robots from 41 companies

7

ON MARKET

Ankon Technologies  
EndoRobotics Co., Ltd.  
Endotics System  
G.I. View Ltd  
Jinshan Science & Technology  
Neptune Medical  
Nitinotes Surgical

27

UNDER DEV

Agilis Robotics  
Aohua Endoscopy  
Aopeng Medical / Operation R...  
Atlas Endoscopy  
Axtes / Aiketes  
Baifu  
Channel Robotics  
EndoQuest Robotics  
EndoR Surgical System  
Ergami Endoscopy  
Huiyue Medical  
HumanTouch Surgical  
INEN Robotics  
InterMag  
Inventor Technology  
Keypoint Surgical  
MedInTech  
NISI  
Radius Dynamics GmbH  
Serpenta  
Swan EndoSurgical, Inc.  
Ultimage Health Technology C...  
Vivo Surgical  
W Endoluminal Robotics  
Weihai Kefan Medical Device ...  
Yunlijing'an Technology  
Yuying Medical International...

2

UNCERTAIN

Boston Scientific  
Fractyl Health

5

DEAD

Invendo Medical / Ambu  
Medrobotics Corporation  
MichelAngelo RobotScope  
Motus GI Medical Technologies  
NeoGuide Systems

# Lung Endoluminal

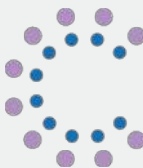
Representative examples



INTUITIVE



GALAXY



Noah Medical

AURIS  
Johnson & Johnson  
MedTech



MONARCH™ QUEST

SPIRO Robotics



# Lung Endoluminal / Bronchial

20 robots from 20 companies

7

ON MARKET

Edge Medical Robotics / 精锋医疗  
Intuitive Surgical  
JNJ MedTech - Monarch  
Langkai Medical  
MicroPort MedBot  
Noah Medical  
Spiro Robotics

10

UNDER DEV

Anteeo Surgical / 北京跃维医疗科技  
Apeio Health Co., Ltd.  
Apreo Health  
Broncus Medical / 堃博医疗  
Hanglok-Tech  
LungHealth Medtech  
Magnisity  
SyncRobotix, Inc.  
Vine Medical, Inc.  
Xuzhou Yintu Medical Researc...

1

ACQUIRED

Auris Health

1

UNCERTAIN

Sanovas, Inc.

1

DEAD

Borns Medical Robotics / 博恩思

# Urology

Representative examples



# Urology

33 robots from 28 companies

16

## ON MARKET

BestMT / 拜斯特医疗科技  
Biobot Surgical  
ELMED Medical Systems  
FocalTherics / EDAP TMS  
Innomedicus / ARTEMIS  
Intuitive Surgical  
Jiehang Robot / 上海介航机器人  
JNJ MedTech - Monarch  
Medicaroid Corporation  
Medtronic  
MicroPort MedBot  
PROCEPT BioRobotics  
Profound Medical  
ROEN Surgical / 로엔서지컬  
Sterlab  
Surgerii Robotics

9

## UNDER DEV

Andromeda Surgical  
Healinno Tech  
Insight Medbotics  
LARC Robotics  
Needleye Robotics Srl  
Promaxo, Inc.  
Richard Wolf GmbH  
Sagebot / 思哲睿  
Virtuoso Surgical

1

## UNCERTAIN

Continuum Therapeutics

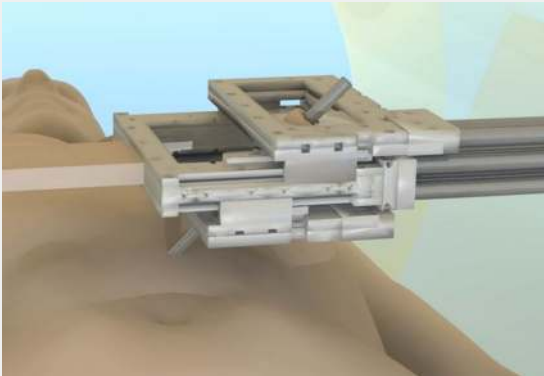
2

## DEAD

Comau SpA  
Polymer Robotics Inc.

# Percutaneous

Representative examples



GUIDABOT, LLC



MENDERA



# Percutaneous

54 robots from 51 companies

20

ON MARKET

CAScination  
Ganaivi Medical / 浙江伽奈维医疗  
HistoSonics  
Insight Medbotics  
Insightec  
Interventional Systems  
LifeTech-Quantum Medical Tec...  
Longdianjing Intelligent Tec...  
MASMEC Biomed  
Mendaera  
NDR Medical Technology  
Perfint Healthcare Pvt Ltd  
Praxis Holding LLC  
Quantum Surgical  
Ruitouch Technology / 上海睿触科技  
Soteria Medical  
Tuodao Medical  
Weide Precision Medical  
Zhenjiankang Medical / TrueH...  
Zhongrui Funing / 中瑞福宁

23

UNDER DEV

AIR Surgical  
AIRS Medical  
Aoda Zhisheng Medical Techno...  
Apodibot Medical  
Baird Medical  
BEC / Guidoo  
Cosmic M.E.  
Cure-in  
DEMCON Medical Robotics  
Edda Technology  
FlexPerc Medical  
Hanglok-Tech  
Hengrui Hongyuan Medical Tec...  
Libang Surgical Technologies...  
LUKE Robotics  
Medical Devices Corner / MDC  
Miaoweishi Technology  
Nawei Medical / 奈维医疗  
Puncture Robotic / 磅策医疗  
Shengdankang Medical Technol...  
Xianyang Robot Co.  
Yida Health Medical Technolo...  
Yingtai Li'ankang

1

ACQUIRED

Innomedic / Synthes / JNJ

1

UNCERTAIN

RobotX Medical

6

DEAD

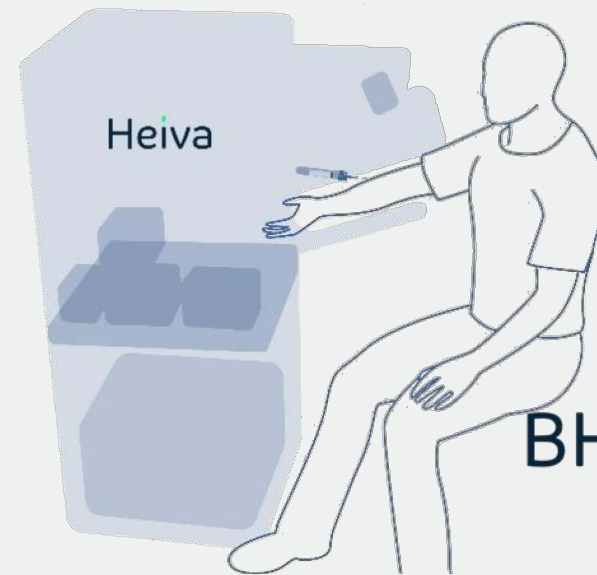
GuidaBot LLC  
Robopsy  
Seiko Epson  
Starpax Biopharma  
XACT Robotics Ltd.  
Yikangtai Medical Technology

# Vascular Access

Representative examples



VITESTRO



BHealthCare



VEEBOT SYSTEMS

# Vascular Access

24 robots from 23 companies

2

## ON MARKET

MagicNurse / 迈纳士  
Vitestro

18

## UNDER DEV

AutonomUS Medical Technologies  
BHealthCare  
Bracco Imaging  
Cobionix  
Create LLC  
Faction Imaging  
Fresenius  
Fuxi Jiuzhen Intelligent Tec...  
Grifols Worldwide Operations...  
Hyperion Surgical  
Kairui Medical / 凯瑞医疗  
Nigale Biomedical / 四川南格尔生物科技  
Obvius Robotics  
Puncture Robotic / 磅策医疗  
Takeda Pharmaceutical / 武田薬品  
Triton Systems  
VascuLogic, LLC  
X9, Inc.

3

## UNCERTAIN

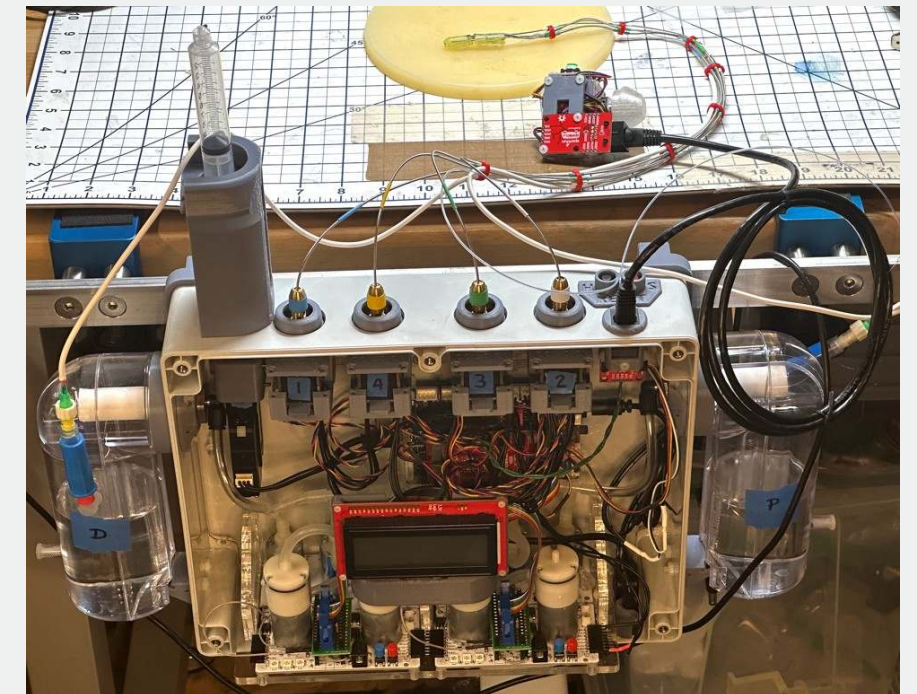
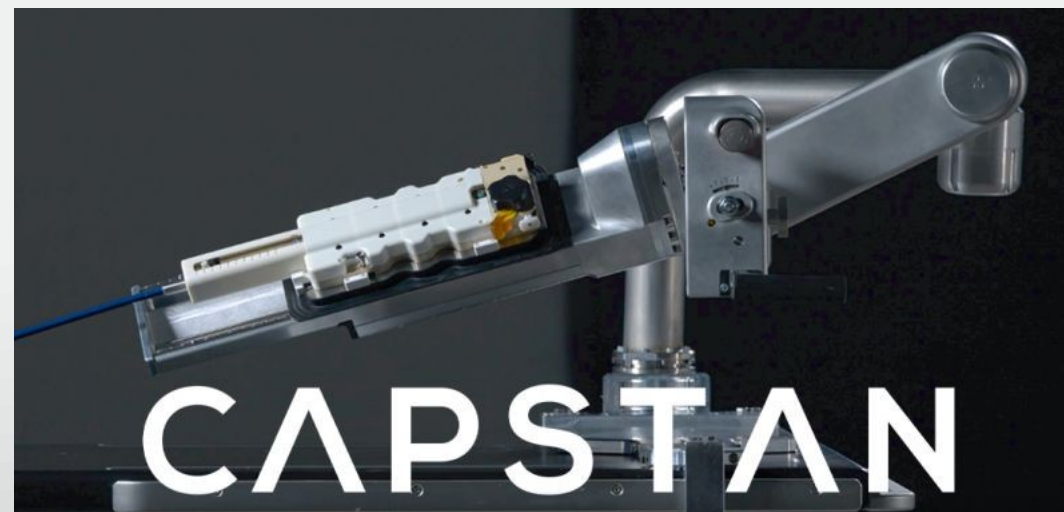
Gliatech SL  
Sheba Impact  
Veebot Systems, Inc.

# Cardiac

Representative examples



**CARANX  
MEDICAL**



# Cardiac

32 robots from 32 companies

5

## ON MARKET

Cardiawave  
LN Robotics  
Mayo Cardiomagnetics / Meio ...  
Stereotaxis, Inc.  
Transmural Systems

19

## UNDER DEV

Biosense Webster  
Capstan Medical  
Conform Medical  
Denuo Electrophysiology Medi...  
DigiSurge  
Edwards Lifesciences  
EndoSurge  
Indian Wells Medical  
Jenscare Scientific  
Laza Medical  
Neuro-Kinesis Corp.  
Peijia Medical / 沛嘉医疗  
Pnemed Technology Limited  
ROB'E GmbH  
Shifamed Holdings  
Surlogic / SurgiPulse Robotics  
Symphony Robotics  
UNandUP  
Zhicheng Medical / 智程医疗

3

## ACQUIRED

Caranx Medical  
CyberHeart / Varian  
VytronUS, Inc.

2

## UNCERTAIN

CardiaTouch Control Systems  
HeartLander

3

## DEAD

Hansen Medical / JNJ  
Moray Medical  
St. Jude Robotic Catheter

# Microsurgery

Representative examples



MMI®



SONY



GALEN ROBOTICS



microsure  
robot assisted microsurgery



# Microsurgery

12 robots from 12 companies

4

## ON MARKET

BHS Technologies  
Lydus Medical  
Medical Microinstruments  
Microsure

6

## UNDER DEV

Deepsurgery Medical Instrument  
Dessight Biomedical  
F.MED / F . M E D 株式会社  
Gibbon Medical  
KouTech / 昂泰微精  
Sony

1

## ACQUIRED

True Digital Surgery

1

## UNCERTAIN

Elytra Technologies

# Ophthalmic

Representative examples



PRECEYES

Acusurgical



FORSIGHT  
ROBOTICS



HORIZON  
SURGICAL SYSTEMS

# Ophthalmic

26 robots from 26 companies

5

ON MARKET

Belkin Vision  
Beyeonics  
LENSAR  
PRECEYES Medical Robotics  
Riverfield Inc.

19

UNDER DEV

Acusurgical  
Alcon  
Dessight Biomedical  
ForSight Robotics  
Horizon Surgical Systems, Inc.  
Impel IP LLC  
J&J Surgical Vision  
Keranova  
Mingche Biotechnology / 明澈生物科技  
Mynutia  
Ocutrx Vision Technologies  
Ophthorobotics AG  
Optorobotics S.r.l.  
RIMSCIENCE Co., Ltd.  
Robotrak  
Suroviz  
TIROMU / 瞳沐医疗  
Weimou Medical / 微眸医疗  
Xianwei Medical Technology

2

DEAD

Medical Robotic Technologies...  
NSK Ltd. / 日本精工

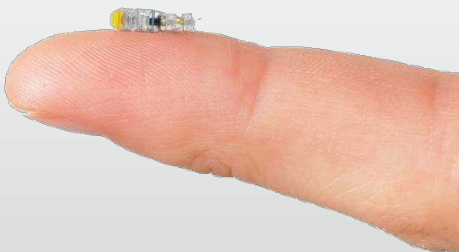
# Micro / Nano Bots

Representative examples

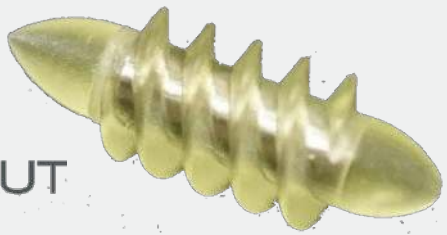


 ARTEDRONE

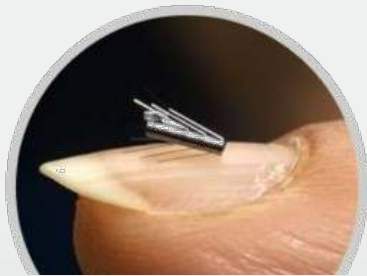
ROBEAUTE



 BIONAUT  
LABS



 microbot  
medical



 ViRob  
Life in motion

# Micro / Nano

12 robots from 11 companies

10

UNDER DEV

Bionaut Labs  
IMSystem / 아임시스템  
InterMag  
Interventional Systems  
Koswire / 코스와이어  
Macheon AI  
MagnebotiX  
Robeauté  
Symphony Robotics  
ValueBioTech S.R.L.

1

ACQUIRED

ArteDrone

# Otolaryngology

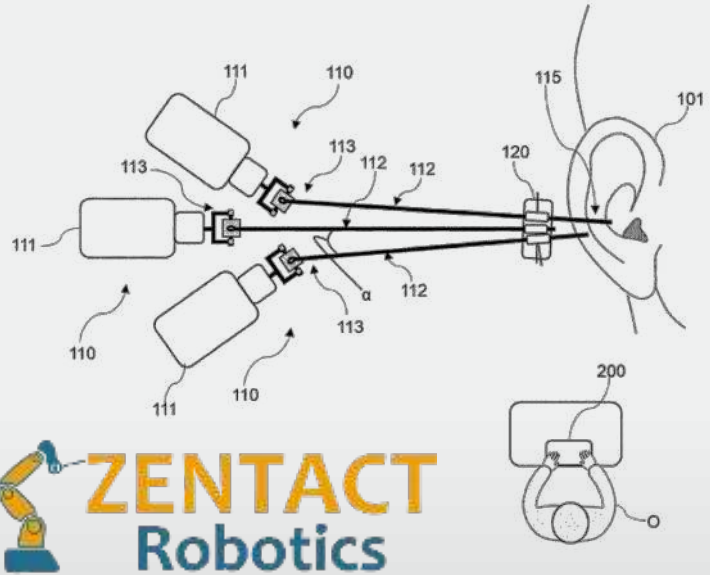
Representative examples



 **Medrobotics**



**iota**



 **ZENTACT**  
Robotics



 **Amal**  
TECHNOLOGIES



 **思哲睿医疗**  
SAGEBOT MEDICAL

# Otolaryngology

18 robots from 17 companies

4

ON MARKET

Collin Medical  
Galen Robotics  
iotaMotion  
MED-EL / CAsCination

10

UNDER DEV

Amarob Technologies  
Borns Medical Robotics / 博恩思  
Cochlear Limited  
NavLab Inc.  
Sagebot / 思哲睿  
Tamar Robotics Ltd.  
Taurean Surgical, Inc.  
Tianchen Medical  
Weigao Group / 威高集团  
Zentact Robotics

3

DEAD

Brain Navi Biotechnology  
InnovoTone LLC  
URS



# Exits, failures, and the money

Who was acquired · who died · who buys · exit economics · SPACs · what changed in 2026 · where the money sits

# Robots or companies whose status changed: acquired

45 companies (46 robot rows) carry Acquired status, Computer Motion (2003) to Robocath (2026)

## A - C

ArteDrone · Asensus Surgical · Auris Health · Auris Health / JNJ · Blue Belt Technologies · Caranx Medical · Cardan Robotics · Computer Motion · Corindus / Siemens Healthineers · Covidien · CyberHeart / Varian

## D - K

Denso Corporation · Endocontrol SA · Excelsius Surgical / Globus Medical · Fusion Robotics / Accelus · Gomtec / ABB · Great Belief International Limited / GBIL · Human Xtensions / Human Extensions Ltd. · Innomedic / Synthes / JNJ · JNJ Ethicon Endo-Surgery RAD program · KAT Robotics · KB Medical

## L - O

LinkX Robotics · Mako Surgical Corp. · Mazor Robotics · MDA · Medical Surgery Technologies · Medineering / Brainlab · Mobius Imaging / Stryker · Mobius Imaging / Stryker - Mako Spine · Monogram Technologies · OMNIlife Science / Orthopedic Synergy Inc. · OrthoSpin Ltd.

## O - V

Orthotaxy SAS · Ostesys · Relievent Medsystems · RoboCath · Ruby Robotics · Sensus Healthcare, Inc. · SOFAR S.p.A. · TransEnterix · True Digital Surgery · Tuge Healthcare · Verb Surgical · VytronUS, Inc.

# Robots or companies whose status changed: dead

63 dead companies/programs (65 robot rows), up from 48 in January, several parents remain active elsewhere

## A - C

Acrobot / Stanmore · Advanced Osteotomy Tools · Aeon Scientific · Agile EndoSurgery · American GNC Corporation · Avatera Medical · AVRA Medical Robotics · Bio-Medical Engineering · Borns Medical Robotics / 博恩思 · Boston Scientific · Brachium · Brain Navi Biotechnology / 钛隼生物科技 · Catheter Precision, Inc. · Centauri Robotic Surgical Systems · Comau SpA

## C - L

CranioMedix Inc. · Cyber-Implants LLC · Euphrates Vascular / Pulse Therapeutics · FocalTherics / EDAP TMS · GuidaBot LLC · Hansen Medical / JNJ · Human Robotics S.A. de C.V. · iin medical · InnovoTone LLC · Intuitive Surgical · Invendo Medical / Ambu · Invivo Robotics · ISIS · Lapara Surgical · LG Electronics · Longhui Medical / 上海龙慧医疗

## M - R

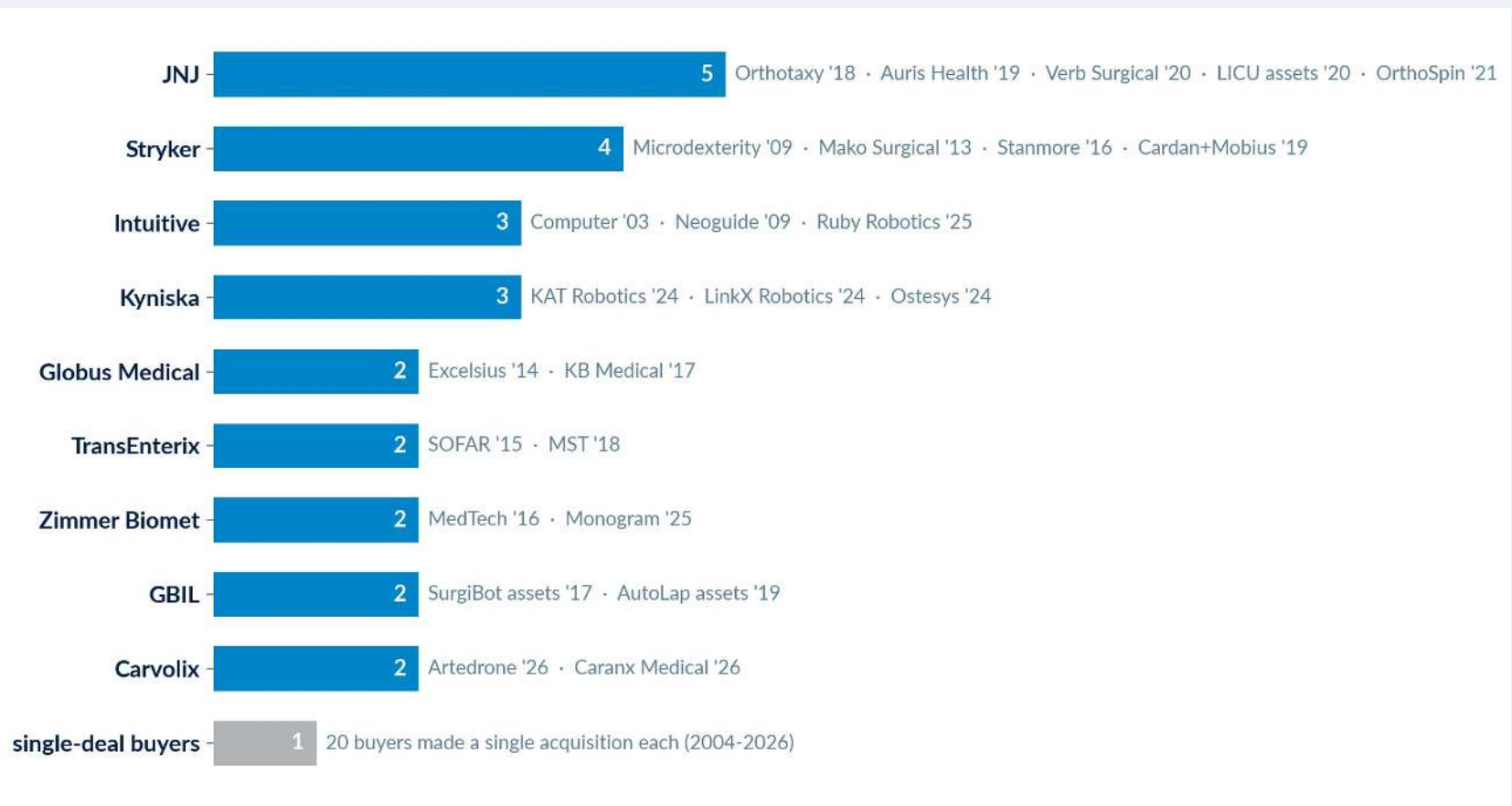
MDA · MDA Space Ltd. · Medical Robotic Technologies B.V. · Medrobotics Corporation · MichelAngelo RobotScope · Moray Medical · Motus GI Medical Technologies · National Semiconductor · NeoGuide Systems · NeuroArm / SYMBIS · NSK Ltd. / 日本精工 · Omniboros · ortoMaquet GmbH / Maquet · Panasonic Holdings / パナソニック · Polymer Robotics Inc. · Revision Robotics Surgical Innovations, LLC

## R - Y

Robopsy · Roboscopy Medical Ltd. · Sagitta Robotics Limited · Samsung Electronics · Seiko Epson · Siemens Healthineers Endovascular Robotics · Signature Robot Ltd · SPI Surgical, Inc. · St. Jude Robotic Catheter · Starpax Biopharma · Synaptive Medical · Telehealthrobotics · URS · Vicarious Surgical Inc. · XACT Robotics Ltd. · Yikangtai Medical Technology

# Exits concentrate in a few repeat buyers

Acquisitions per buyer with targets and deal years; single-deal buyers aggregated. Sources: deal press releases and SEC filings; collected 8 Jul 2026



## 45 RESEARCHED DEALS

**9 repeat buyers account for 25 of 45 acquisitions**

Busiest: JNJ, Stryker, Intuitive, Kyniska

**Repeat buying concentrates in orthopedics/spine and soft tissue**

# Acquisition typically takes about a decade

Age at acquisition vs deal year. Sources: press releases and SEC filings; current as of 8 Jul 2026



## MEDIAN AGE AT EXIT: 9 YEARS

**45 acquisitions with researched terms, 2003 to 2026**

Disclosed prices span \$5M to \$3.4B; billion-dollar deals labeled

**Gray dots: undisclosed price. Asset deals and mergers carry \* in the table**

# Exit economics: the full record (1 of 2)

Raise-to-exit economics, UD = undisclosed, \* = special aspects; maintained in the deck data file; current as of 8 Jul 2026

| COMPANIES                                | FOCUS AREA                | COUNTRY     | INVESTMENT(\$M)        | # ROUNDS | ACQ PRICE (\$M)        | ACQ YEAR | AGE | # INVESTORS | ACQUIRER                |
|------------------------------------------|---------------------------|-------------|------------------------|----------|------------------------|----------|-----|-------------|-------------------------|
| ACROBOT                                  | orthopedic                | UK          | 5                      | 1        | UD                     | 2010     | ~5  | 3           | Stanmore                |
| ARTEDRONE*                               | neurovascular             | France      | ~23                    | UD       | ~12.5                  | 2026     | 9   | 1           | Carvolix                |
| ASENSUS                                  | abdominal/thoracic        | US          | 484                    | 18       | 95                     | 2024     | 18  | 12          | Storz                   |
| AURIS HEALTH*                            | abdominal/thoracic        | US          | 733                    | 8        | 3400                   | 2019     | 12  | 13          | JNJ                     |
| AVATERA MEDICAL*                         | abdominal/thoracic        | Germany     | ~270                   | UD       | UD                     | 2023     | 9   | 1           | Tennor International AG |
| BLUE BELT TECHNOLOGIES*                  | orthopedic                | US          | 45                     | 2        | 275                    | 2015     | 12  | 4           | Smith&Nephew            |
| CARANX MEDICAL*                          | cardiovascular            | France      | ~25 + 5 grants         | UD       | ~18                    | 2026     | 8   | 2+          | Carvolix                |
| CARDAN ROBOTICS + MOBIUS IMAGING*        | spine                     | US          | 45 + ~82               | 2        | 500                    | 2019     | 11  | 1           | Stryker                 |
| COMPUTER MOTION*                         | abdominal/thoracic        | US          | 37(IPO) + 22+ post-IPO | ?        | ~150                   | 2003     | 14  | ?           | Intuitive               |
| CORINDUS                                 | endovascular              | US          | 134                    | 7        | 1100                   | 2019     | 18  | 4           | Siemens                 |
| CYBERHEART                               | cardiac                   | US          | 24                     | 4        | UD                     | 2019     | 13  | 3           | Varian                  |
| DENSO / IARMS*                           | surgeon arm-support robot | Japan       | n/a                    | n/a      | UD                     | 2018     | n/a | n/a         | Toho Technology         |
| ENDOCONTROL                              | abdominal/thoracic        | France      | 6                      | 1        | UD                     | 2019     | 13  | 2           | Canady Robotics         |
| ENDOVIA MEDICAL                          | abdominal/thoracic        | US          | 12                     | 4        | 5                      | 2004     | 6   | 3?          | Hansen Medical          |
| EXCELSIUS SURGICAL*                      | spine                     | US          | 5.5                    | 1        | UD                     | 2014     | 3   | ?           | Globus                  |
| FUSION ROBOTICS / ACCELUS / REMI ASSETS* | spine                     | USA         | UD                     | UD       | 55                     | 2023     | 4   | UD          | ATEC                    |
| GOMTEC*                                  | abdominal/thoracic        | Germany     | UD                     | UD       | UD                     | 2015     | 7   | UD          | ABB                     |
| HANSEN MEDICAL                           | endovascular              | US          | 116                    | 5        | 240(IPO) + 80          | 2016     | 13  | 5           | Auris                   |
| INNOMEDIC*                               | orthopedic                | Germany     | ?                      | ?        | UD                     | 2008     | ~8  | ?           | Synthes                 |
| KAT ROBOTICS*                            | orthopedic                | France      | ~0.5                   | 1        | UD                     | 2024     | 2   | 1           | Kyniska Robotics        |
| KB MEDICAL*                              | spine                     | Switzerland | 7                      | 4        | ~31.5 + 4.9 contingent | 2017     | 5   | 8           | Globus                  |
| LINKX ROBOTICS*                          | orthopedic                | France      | ~0.5                   | 1        | UD                     | 2024     | 3   | 1           | Kyniska Robotics        |
| MAKO SURGICAL*                           | orthopedic                | US          | 95                     | 5        | 51(IPO) + 1700         | 2013     | 9   | 7           | Stryker                 |

## WHAT THE RECORD SHOWS

24 of 45 deals disclose a price; median \$79M

4 deals of \$1B+ sit atop a long tail: most exits are under \$100M

20 undisclosed deals cluster in the early years and small tuck-ins

7 asset-level deals (SurgiBot, AutoLap, REMI, Sculptura) show IP outliving its company

Spine and orthopedics account for the largest share of exits

UD = undisclosed · \* = special aspects (see data file) · IPO = initial public offering · CVR = contingent value right

# Exit economics: the full record (2 of 2)

Raise-to-exit economics, UD = undisclosed, \* = special aspects; maintained in the deck data file; current as of 8 Jul 2026

| COMPANIES                                | FOCUS AREA                                                           | COUNTRY | INVESTMENT(\$M)          | # ROUNDS | ACQ PRICE (\$M) | ACQ YEAR | AGE    | # INVESTORS     | ACQUIRER                 |
|------------------------------------------|----------------------------------------------------------------------|---------|--------------------------|----------|-----------------|----------|--------|-----------------|--------------------------|
| MAZOR SURGICAL TECH                      | spine                                                                | Israel  | 70                       | 3        | 1700            | 2018     | 17     | 6               | Medtronic                |
| MEDICAL SURGERY TECH*                    | abdominal/thoracic                                                   | Israel  | 14                       | 2        | ~31             | 2018     | 13     | 6               | Transenterix             |
| MEDINEERING*                             | neuro                                                                | Germany | UD                       | 2        | UD              | 2019     | 5      | 4               | Brainlab                 |
| MEDTECH                                  | neuro & spine                                                        | France  | 2                        | 1        | 188             | 2016     | 14     | 1               | Zimmer                   |
| MICRODEXTERITY*                          | orthopedic                                                           | US      | UD                       | ?        | UD              | 2009     | 17     | ?               | Stryker                  |
| MONOGRAM TECHNOLOGIES*                   | orthopedic                                                           | USA     | ~75 + 30(IPO) + post-IPO | 10       | 177 + CVR       | 2025     | 9      | crowd + Pro-Dex | Zimmer Biomet            |
| NEOGUIDE                                 | gastroenterology                                                     | US      | 39                       | 2        | UD              | 2009     | 9      | 9               | Intuitive                |
| OMNILIFE SCIENCE                         | orthopedic                                                           | US      | 32                       | 3        | UD              | 2019     | 20     | 2               | Corin Group              |
| ORTHOSPIN*                               | orthopedic                                                           | Israel  | 8                        | ~3       | 79.5            | 2021     | 7      | 2               | DePuy Synthes            |
| ORTHOTAXY*                               | orthopedic                                                           | France  | ?                        | ?        | UD              | 2018     | 9      | ?               | JNJ                      |
| OSTESYS*                                 | orthopedic                                                           | France  | UD                       | UD       | UD              | 2024     | 15     | 2               | Kyniska Robotics         |
| RESTORATION ROBOTICS*                    | hair                                                                 | US      | 129                      | 9        | 25(IPO) + stock | 2019     | 17     | 5               | Venus Concepts           |
| ROBOCATH*                                | vascular intervention                                                | France  | ~88                      | 6        | 20 + 25 earnout | 2026     | 17     | 10+             | Stereotaxis              |
| RUBY ROBOTICS*                           | autonomous point-of-care<br>biopsy robotics + tissue-<br>adequacy AI | USA     | 5.6                      | ~2       | UD              | 2025     | 4      | 4               | Intuitive Surgical       |
| SENSUS HEALTHCARE /<br>SCULPTURA ASSETS* | robotic IORT / brachytherapy<br>radiation oncology                   | USA     | n/a                      | n/a      | 15              | 2022     | n/a    | n/a             | Empyrean Medical Systems |
| SO FAR*                                  | abdominal/thoracic                                                   | Italy   | UD                       | ?        | 100             | 2015     | ~7     | 1               | Transenterix             |
| STANMORE IMPLANTS*                       | orthopedic                                                           | UK      | 34                       | ?        | 52              | 2016     | 6 / 68 | ?               | Stryker                  |
| TRANSENTERIX / AUTOLAP<br>ASSETS*        | abdominal                                                            | USA     | n/a                      | n/a      | 47              | 2019     | n/a    | n/a             | GBIL                     |
| TRANSENTERIX / SURGIBOT<br>ASSETS*       | abdominal                                                            | USA     | n/a                      | n/a      | 29              | 2017     | n/a    | n/a             | GBIL                     |
| TRUE DIGITAL SURGERY*                    | digital surgical microscopy<br>robotics                              | USA     | ~49                      | UD       | UD              | 2025     | 7      | 5               | B. Braun                 |
| VERB SURGICAL *                          | abdominal/thoracic                                                   | US      | ~500                     | 1        | UD              | 2020     | 5      | 2               | JNJ                      |
| VYTRONUS / LICU ASSETS*                  | cardiac EP                                                           | US      | ~115                     | 7        | UD              | 2020     | 14     | 5               | JNJ                      |

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# Special Purpose Acquisition Corporations (SPACs)

The SPAC cohort, and how it went: Memic SPAC failed, re-emerged as Momentis; Vicarious delisted Mar 2026, board dissolution vote set



debut on Wall Street on Monday. The company closed its previously announced business combination with D8 Holdings Corp. last week, gathering \$220 million in gross proceeds. The deal, announced this past April, valued the combined company at an enterprise value of approximately \$1.1 billion.



Israel medical device company **Memic Innovative Surgery Ltd.** today announced the signing of a definitive agreement for a SPAC merger with MedTech Acquisition Corporation (Nasdaq: MTAC). The merged company will have \$360 million cash and an equity value of more than \$1 billion.

# Robots or companies whose status changed in 2026

Exits validate the category; the flameouts show that approval  $\neq$  commercialization

## EXITS & WINS

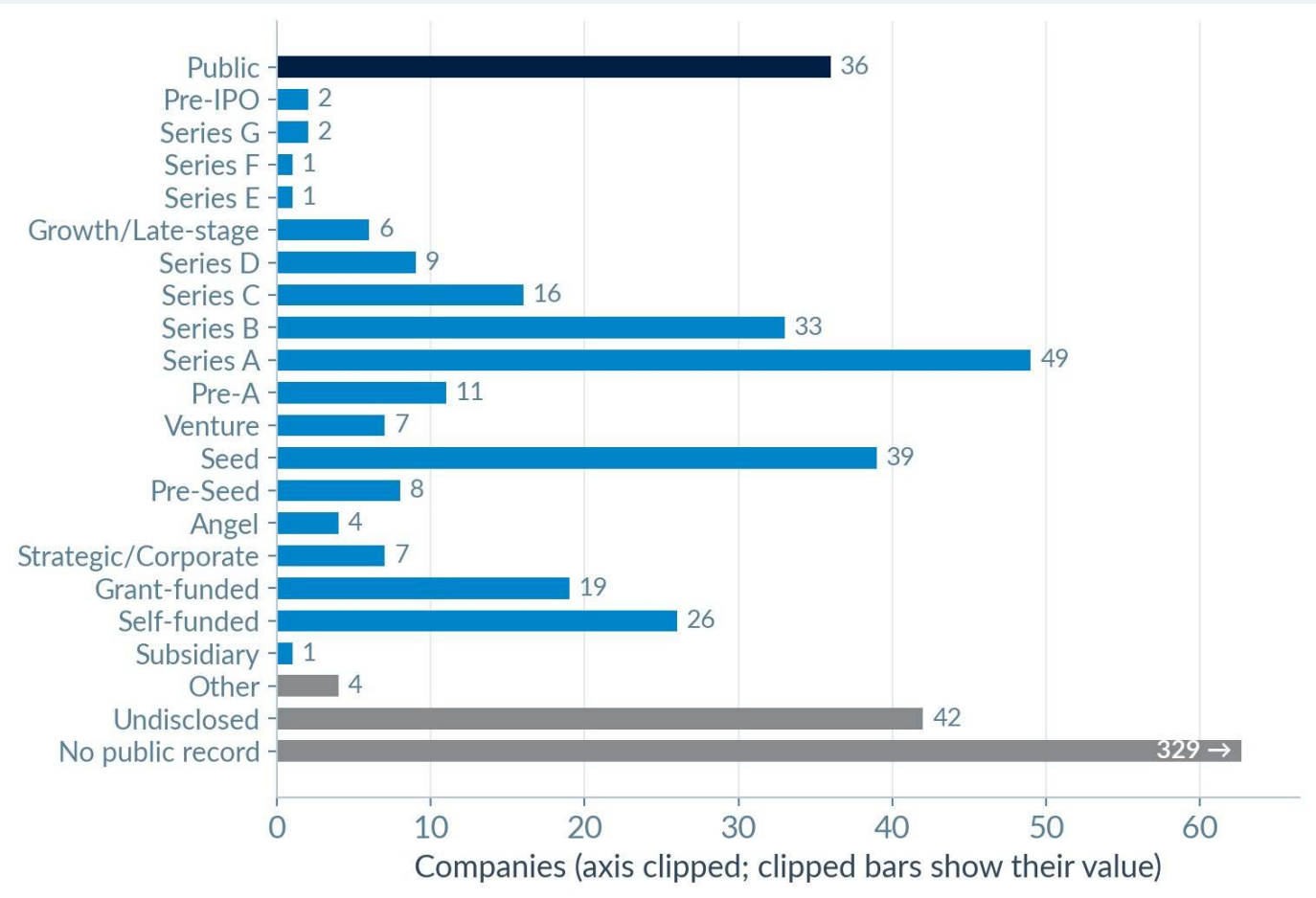
- **TrueHealth / 真健康 → HKEX IPO (Jun 30)**  
+217% day one, ~HK\$14.3B cap, ~1,800× oversubscribed, capital is concentrating on the category leader
- **Edge Medical / 精锋 → Hang Seng Composite**  
Index inclusion Jun 8; FY2025 revenue RMB 456M (+185%); overseas installs now exceed domestic
- **Stereotaxis × Robocath**  
Endovascular consolidation (Apr 2026)
- **Ruichu Robotics → HKEX A1 filed (May 27)**  
Pre-revenue lung-puncture robot chasing the TrueHealth window

## WOUND DOWN / DELISTED

- **Vicarious Surgical (US)**  
NYSE delisting Mar 3 (cap <\$15M) → OTC as RBOT; first 2026 Western flameout; dissolved 21 Jul 2026 (board-approved liquidation)
- **Asensus / Senhance + Luna (US/DE)**  
Karl Storz discontinues both robots and retires the brand, a ~20-year program ends (strategic wind-down, not bankruptcy)
- **Longhui Medical / 龙慧 (China)**  
NMPA approval Sep 2024 → bankruptcy Jan 2026: approval  $\neq$  commercialization
- **Microya / 微亚 (China)**  
Entire IP + assets auctioned from ~RMB 1.29M starting bid

# Over half the field has no public funding story

Funding stage per company (normalized), 652 active companies; 11 acquired companies excluded (covered in the exits section)



## 329 NO PUBLIC FUNDING RECORD

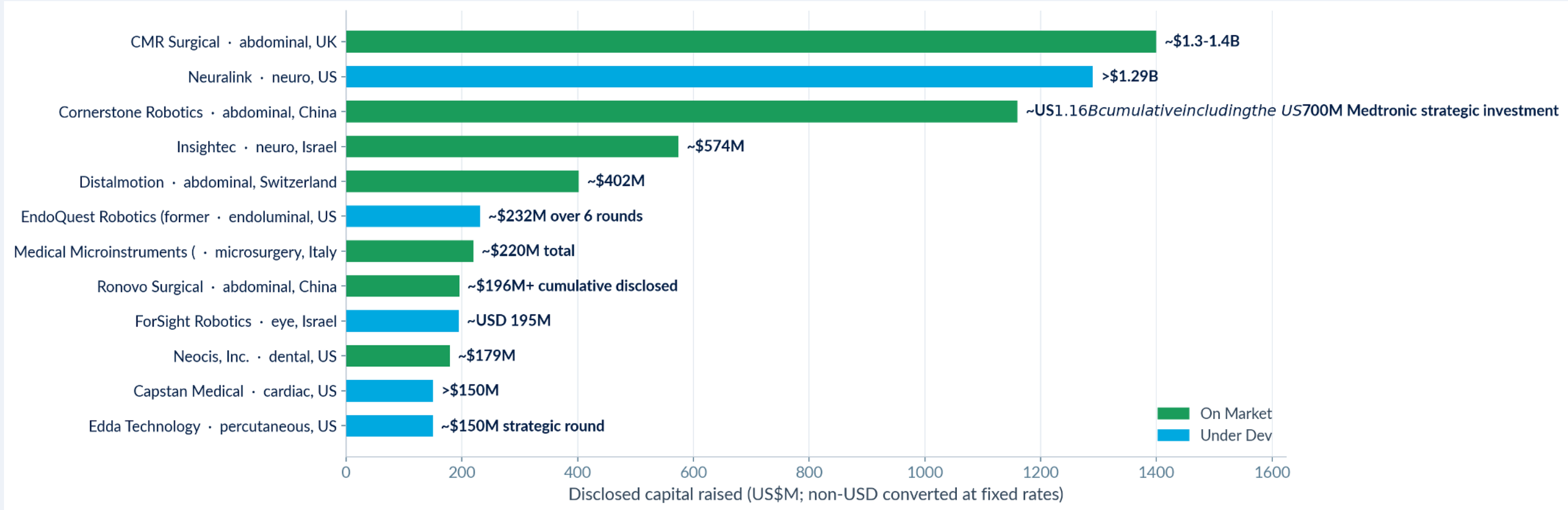
**50% of companies; mostly Chinese patent-only finds, institutes, and quiet SMEs**

36 public companies bookend the other extreme

**161 companies disclose a raise amount; Seed-to-B dominates the private tail**

# Best-funded private players

Ranked by disclosed capital raised (July 2026), per the master roster



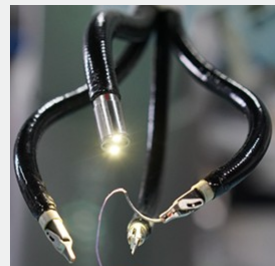


# Examples of insights from assessing a range of commercial efforts

What the collection teaches: form factors, topologies, consoles, controls

# Abdominal/thoracic form factors

Representative examples, single-port remains the exception

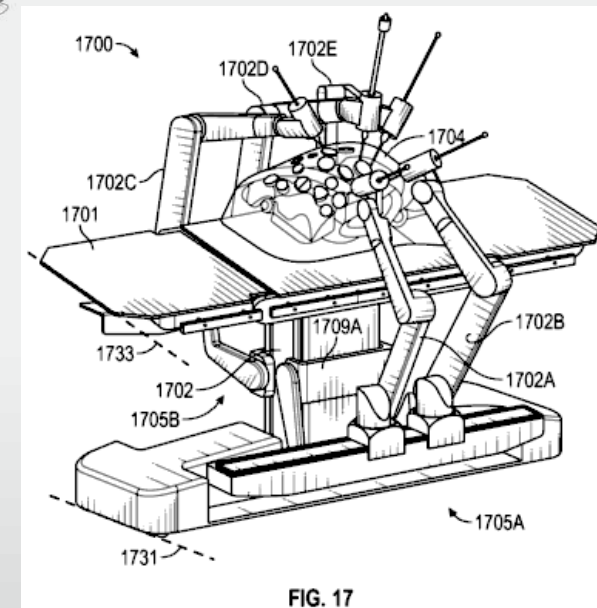
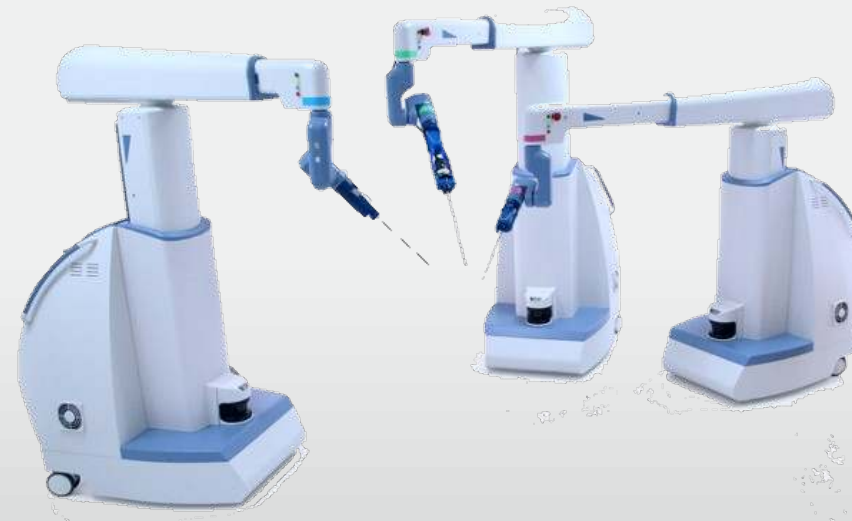
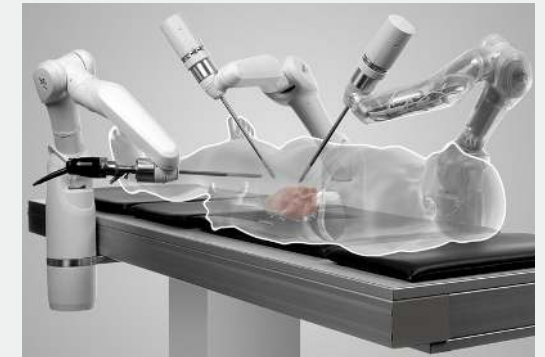


15 Single port

116 Multi port

# Varying topologies

Representative examples



# Varying consoles

Representative examples



INTUITIVE



CMR  
SURGICAL



revo<sup>™</sup>  
surgical solution



TITAN MEDICAL



AESENSUS  
SURGICAL



MicroPort

SS INNOVATIONS



Medtronic



Medicaroid



Distalmotion



avatera



# Varying hand and foot controls

Representative examples



INTUITIVE



CMR  
SURGICAL



Medtronic



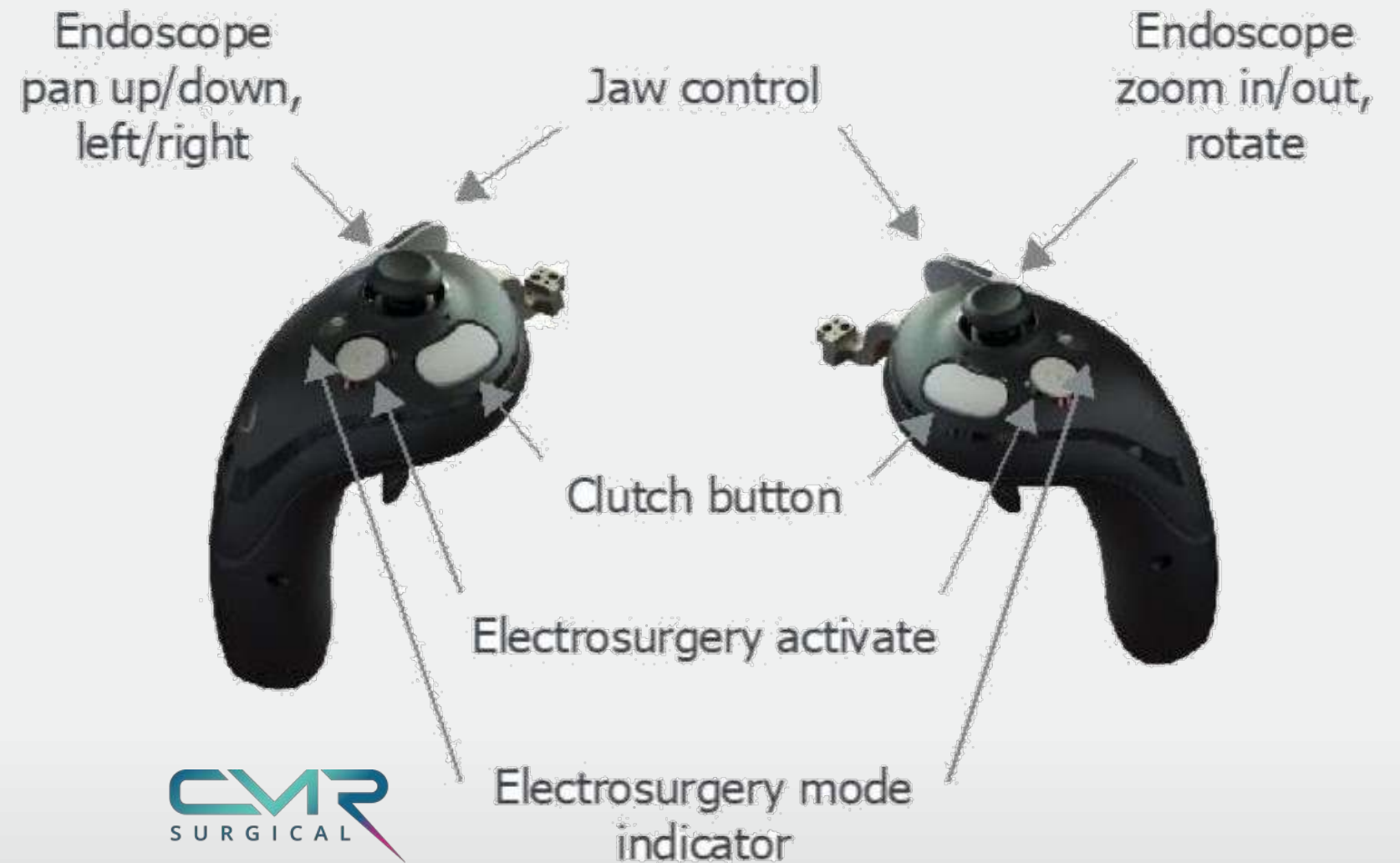
TITAN MEDICAL



Distalmotion

# Simple squeeze vs complex handle

Those missing controls from the squeeze handle live somewhere...



# Novel approaches

Representative examples



# Major trends

Where the field is stuck, and where it should go next

- **Still too many robots in the same specialties**  
...with similar approaches, and still too many full-platform solutions
- **Control-interface variety is a problem**  
Physical and visual control interfaces differ widely from platform to platform
- **Just roboticizing what is done by hand is not enough**  
First autonomous execution of a procedure phase supports this (Science Robotics, 2025)

## GOING FORWARD

Opportunity for existing platform enhancement

- Operate in smaller and/or lower-resourced facilities
- Pay-per-use rather than capital + tool/drape purchases
- Lower capital cost · smaller footprints
- Wear-based rather than number-of-uses-based tool replacement
- Higher-power-density small actuators will change device configurations
- Judicious use of soft actuators as well as live hinges
- Legal frameworks and monetization of telesurgery

# Machine learning

Everywhere, expected, penalized if absent; marketing still runs ahead of reality

- **The barrier to entry keeps dropping**

Technical and data-availability thresholds continue to fall; foundation models increasingly sit beneath specialist data

- **Most applications will be under the hood**

Optimizing the robot itself; more small, inexpensive sensors coupled with ML/AI video

- **Simulation for training content**

There is a strong case for simulation in creating training data and content

## LEVELING OUTCOMES ACROSS LOCATION & SURGEON

Enhanced surgical-site situational awareness

- Contextualization relative to best practices
- Ready availability of ad-hoc mentorship
- Subtask enhancement / automation
- Removing the need for technicians to run things

- Autonomous cholecystectomy phase demonstrated ex vivo, 8/8 (Kim et al., Science Robotics 2025)

*Tools like Claude Code are accelerating and broadening use cases, but will need to be thorough in verification: they very quickly make things that look great but have unclear provenance*

# Surgical Robotics Watch, a live companion dashboard

An updating web version of the data underlying this deck, search, filter, sort, and drill into any company

Surgical Robotics Watch

The full surgical-robotics roster — companies and the robotic systems they build, with configuration, control mode, funding, founding year, status and provenance. Companion to the Surgical AI Watch. Tap any company for full detail.

648 companies745 robots264 on market330 under dev216 US · 223 China · 46 France

WHAT'S CHANGED · 36 UPDATES

▶ 2026-07-06 — 1 changed (745 total)

▶ Earlier updates (35)

Search company, robot, invesAll FocusAll ConfigurationAll ControlAll StatusAll CountryAll Funding stageReset

745 of 745

| COMPANY                                                                                                                                       | FOCUS              | CONTROL    | STATUS    | COUNTRY | FOUNDED | FUNDING      |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------|-----------|---------|---------|--------------|
| ▶ <b>8i Robotics</b><br>Three-arm spine surgery robot (imaging head, ultrasonic cutter, screwdriver) with EM navigation; founded by ex...     | spine              | teleop     | Under Dev | Canada  | 2022    | Undisclosed  |
| ▶ <b>Accrea Medical Robotics</b><br>AVROS-P: scope-holder robot for abdominal/thoracic MIS; part of the AVROS surgical-robot family; early... | abdominal/thoracic | teleop     | Under Dev | Poland  | 2007    | Grant-funded |
| ▶ <b>Accuray</b><br>CyberKnife S7: robotic-arm-mounted linear accelerator delivering image-guided stereotactic...                             | radiation          | preplanned | On Market | US      | 1990    | Public       |
| ▶ <b>Accuray</b><br>Helix: CT-guided helical radiation-therapy system (gantry-based) aimed at emerging markets                                | radiation          | preplanned | On Market | US      | 1990    | Public       |
| ▶ <b>Accuray</b><br>Radixact/SynC: helical TomoTherapy radiation-delivery platform; SynC adds adaptive radiotherapy                           | radiation          | preplanned | On Market | US      | 1990    | Public       |
| ▶ <b>Accuray</b><br>Tomo C: helical radiation-therapy system tailored to China and other emerging markets                                     | radiation          | preplanned | On Market | US      | 1990    | Public       |
| ▶ <b>Accuthera Inc. / 株式会社アキュセラ</b><br>CygneX I: CyberKnife-style robotic-arm radiotherapy system carrying a compact X-band linac             | radiation          | preplanned | Uncertain | Japan   | 2005    |              |
| ▶ <b>Acrobot / Stanmore</b><br>Sculptor: early 'active-constraint' hands-on knee surgery robot; assets absorbed into MAKO/Stryker             | orthopedic         | teleop     | DEAD      | UK      | 1999    |              |
| ▶ <b>Actuated Medical</b><br>GripTract-GI: single-use motorized endoscope-cap 'third-hand' that manipulates tissue during GI...               | gastroenterology   | teleop     | On Market | US      | 2006    | Grant-funded |
| ▶ <b>Acusurgical</b><br>Luca: bi-manual teleoperated robot for vitreoretinal eye                                                              | eye                | teleop     | Under Dev | France  | 2020    | Series A     |

LIVE AT  
[dmeglan.github.io/surgical-robot-companies-dashboard](https://dmeglan.github.io/surgical-robot-companies-dashboard)



## WHAT IT IS

A self-contained, mobile-responsive dashboard generated from the same master spreadsheet behind this deck

## WHAT YOU CAN DO

- Full-text search across company, robot, investor, milestone
- Filter by focus area, configuration, control, status, country, funding stage
- Sortable table; tap any company for full detail with milestones and provenance
- STAYS CURRENT

**Rebuilt with every roster refresh (twice-weekly watch); a "what changed" panel accompanies every update**



# Questions / Discussion?